

10/10/2025 – Hyosung Tx Report (SN:F0247169-T1-0001) at Powerlink Narangba

Client	Hyosung	Narangba Powerlink TX	
Execution date	9/10/2025	Reason of the job	Commission
Tested by	Ben R / Paul A	Location	Hyosung Transformer
Approved by	Cory Hay	Asset	Transformer
Report ID	3	Asset type	Two-winding
Report issue date	14/10/2025 9:41:45 PM	Asset serial number	F0247169-T1-0001
Work order		Manufacturer	HYOSUNG

Summary

Appears to be an issue with Primary bushing B, causing winding DF and CAP to fail and the Omicron Tetrano 600 and CP TD12 not being able to output on that phase while testing DF and CAP of that bushing.

Performed tests	Assessment
Winding DF & CAP	Fail 
Bushing Prim DF & CAP - C1	Investigate 
Bushing Prim DF & CAP - C2	Fail 
Bushing Sec DF & CAP - C1	Pass 
Bushing Sec DF & CAP - C2	Pass 
SFRA Test Tap 1	
SFRA Test Tap 9	
SFRA Test Tap 17	
Exciting Current	Manual pass 
Exciting Current Test After Demag	Manual pass 
Ratio Prim-Sec	Not rated 
DC Winding Resistance Prim	Pass 
DC Winding Resistance Sec	Pass 
Demagnetization	Manual pass 

Overall Assessment	Values conspicuous
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Tested by:
Benjamin Rathbone, Paul Allard

Approved by:
Cory Hay

Location & company information			
Location		Company	
Name	Hyosung Transformer	Company	Yurkia
Region		Department	E & I
Division	Yurika Test	Address	21a Charles Ulm Ave
Area		City	Brisbane
Plant	Powerlink Spare Power Transformer	State/Province	Queensland
Address	66-78 Saltwater Court	Postal code	4009
City	Brisbane	Country	Australia
State/Province	Queensland	Phone No.	
Postal code	4504	Fax No.	
Country	Australia	E-mail	

Geo coordinates	
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Contact person	
Name	Yurika Test
Phone No. 1	
Phone No. 2	
Fax No.	
E-mail	

Comment

Transformer nameplate data			
Serial number	F0247169-T1-0001	Apparatus ID	
Manufacturer	HYOSUNG	Feeder	
Manufacturing year	2025	No. phases	3
Manufacturer type	Power Transformer	Vector group	YNd1

Comment

Voltage ratings			
Winding	Voltage L-L	Voltage L-N	Insul. level L-L
Prim	275.000 kV	158.771 kV	kV
Sec	33.000 kV	kV	kV

Power ratings		
Rated power	Cooling class	Temp. rise wind.
200.000 MVA	ONAN	
250.000 MVA	ONAF	

Current ratings at rated power		
Winding		
Prim	Sec	Rated power
420.000 A	3499.000 A	200.000 MVA
525.000 A	4374.000 A	250.000 MVA

Short-circuit rating		
Max. short-circuit current	40.000 kA	2.0 s

Impedances		
Ref. temp.	75 °C	

Short-circuit impedance Prim - Sec					
uk[%]	Base power	Base voltage	Load losses Pk	OLTC position	DETC position
13.280 %	250.000 MVA	275.000 kV	W		

Zero sequence impedance		
Base power	250.00 MVA	
Base voltage	275.000 kV	
Description	Z0[%]	
Primary	13.28 %	

Others	
Category	Power
Status	Spare
Tank type	Sealed conservator
Insulation medium	Mineral oil
Fluid insulation volume	l
Fluid insulation weight	67900 kg
Total weight	278160 kg

Bushings nameplate data							
Pos.	Serial No.	Manufact.	Manufacturer type	Man. year	Voltage L-ground	Max. system voltage	Rated current
A	313162	HSP	SESTFt 1050-300-B E6 B	2024	158.8 kV	300.0 kV	2000.0 A
B	313160	HSP	SESTFt 1050-300-B E6 B	2024	158.8 kV	300.0 kV	2000.0 A
C	313161	HSP	SESTFt 1050-300-B E6 B	2024	158.8 kV	300.0 kV	2000.0 A
N	313163	HSP	STARTIS-Si+ 52-2500-B E3 B	2024	52.0 kV	52.0 kV	2500.0 A
a1	2422461	Trench High Voltage Products	ECTA- 40.5/6300-4	2024	33.0 kV	33.0 kV	6300.0 A
b1	2402462	Trench High Voltage Products	ECTA- 40.5/6300-4	2024	33.0 kV	33.0 kV	6300.0 A
c1	2402460	Trench High Voltage Products	ECTA- 40.5/6300-4	2024	33.0 kV	33.0 kV	6300.0 A

Pos.	Insul. level LL	Cap. (C1)	DF (C1)	Cap. (C2)	DF (C2)	Insulation type	Tap type
A	1.05 MV	533.00 pF	0.210 %	348.00 pF	%	Resin-impregnated paper	With test tap
B	1.05 MV	540.00 pF	0.220 %	334.00 pF	%	Resin-impregnated paper	With test tap
C	1.05 MV	537.00 pF	0.210 %	341.00 pF	%	Resin-impregnated paper	With test tap
N	250 kV	445.00 pF	0.310 %	201.00 pF	%	Resin-impregnated paper	With test tap
a1	263 kV	1527.00 pF	0.350 %	pF	%	Resin-impregnated paper	With test tap
b1	263 kV	1540.00 pF	0.370 %	pF	%	Resin-impregnated paper	With test tap
c1	263 kV	1528.00 pF	0.340 %	pF	%	Resin-impregnated paper	With test tap

Tap Changers nameplate data

OLTC

Serial number	1ZSC8746511
Manufacturer	ABB
Manufacturer type	VUCGRN 370/600/C
Winding	Prim
Tap scheme	1...33
No. of taps	17

Tap	Voltage
1	302500.0 V
2	299063.0 V
3	295625.0 V
4	292188.0 V
5	288750.0 V
6	285313.0 V
7	281875.0 V
8	278438.0 V
9	275000.0 V
10	271563.0 V
11	268125.0 V
12	264688.0 V
13	261250.0 V
14	257813.0 V
15	254375.0 V
16	250938.0 V
17	247500.0 V

Test set information

Model	Serial number	Calibration date
TESTRANO 600	GN096T	2023-08-07
CP TD12	CN732U	2022-07-30

Global test conditions

Weather	Clear	Humidity	76 %
Unit location		Ambient temperature	27 °C

Winding DF & CAP					
Ambient temperature		29 °C		Weather	Clear
Top oil temperature		30 °C		Humidity	50 %

Comments

Assessments cause concern, appears to be due to an issue with primary bushing B, as noticed during the tan δ of primary bushing B.

Standard test

Block 1: injection at Prim

Corr. temperature		30 °C								
Corr. factor		0.8								
* Reference voltage		10.0 kV								
No.	Meas.	Test mode	Freq.	V out	* I out	* Watt losses	DF meas	DF corr	Cap. meas	Assessment
1	ICH+ICHL	GST	50.00 Hz	10.00 kV	42.58 mA	1960.91 mW	0.4605 %	0.3684 %	13552.1 pF	Pass
2	ICH	GSTg-A	50.00 Hz	10.00 kV	18.83 mA	2129.69 mW	1.1308 %	0.9046 %	5991.6 pF	Inv.
3	ICHL	UST-A	50.00 Hz	10.00 kV	23.76 mA	-132.21 mW	-0.0556 %	-0.0445 %	7563.4 pF	Fail
Cross check: calculated ICHL					23.75 mA	-168.78 mW	-0.0711 %		7560.5 pF	Fail

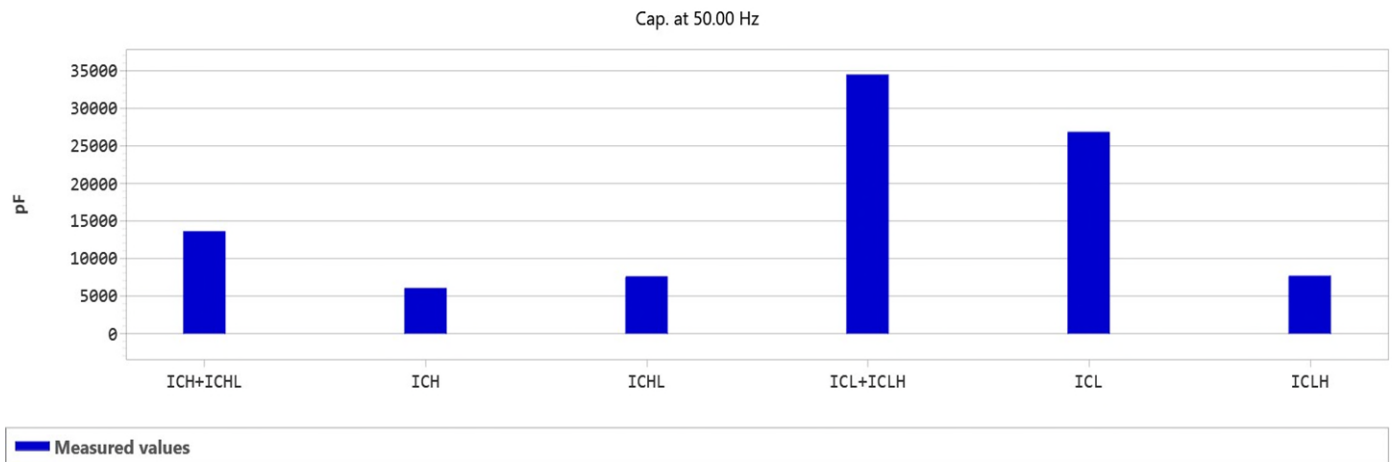
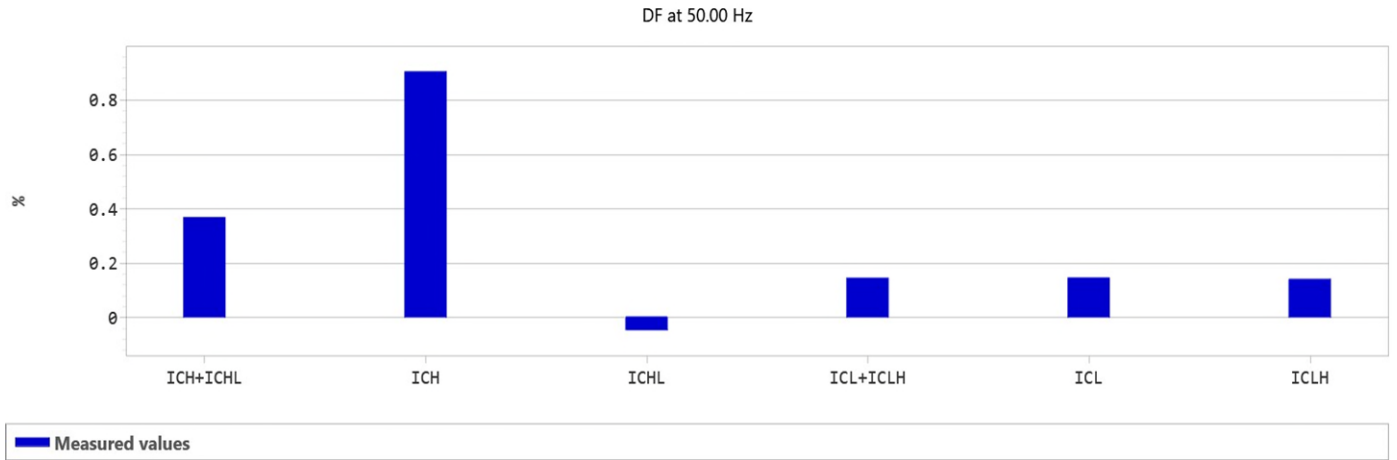
Block 2: injection at Sec

Corr. temperature		30 °C								
Corr. factor		0.8								
* Reference voltage		10.0 kV								
No.	Meas.	Test mode	Freq.	V out	* I out	* Watt losses	DF meas	DF corr	Cap. meas	Assessment
4	ICL+ICLH	GST	50.00 Hz	10.00 kV	108.13 mA	1958.27 mW	0.1811 %	0.1449 %	34416.9 pF	Pass
5	ICL	GSTg-A	50.00 Hz	10.00 kV	84.19 mA	1540.42 mW	0.1830 %	0.1464 %	26796.8 pF	Pass
6	ICLH	UST-A	50.00 Hz	10.00 kV	23.94 mA	421.67 mW	0.1761 %	0.1409 %	7619.9 pF	Pass
Cross check: calculated ICLH					23.94 mA	417.85 mW	0.1745 %		7620.1 pF	Pass

Calculated cross checks

* Reference voltage		10.0 kV			
Test	* I out	* Watt losses	DF meas	Cap.	Assessment
ICHL	23.76 mA	-132.21 mW	-0.0556 %	7563.4 pF	N. rated
ICLH	23.94 mA	421.67 mW	0.1761 %	7619.9 pF	

Graphs for standard test



Bushing Prim DF & CAP - C1

Ambient temperature	24 °C	Weather	Clear
Top oil temperature	24 °C	Humidity	51 %

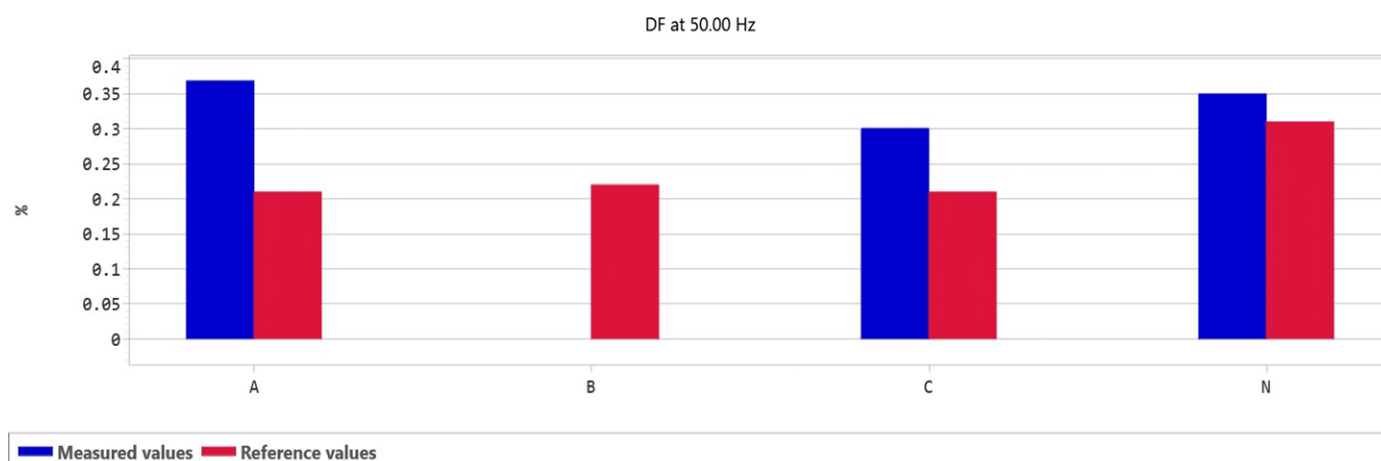
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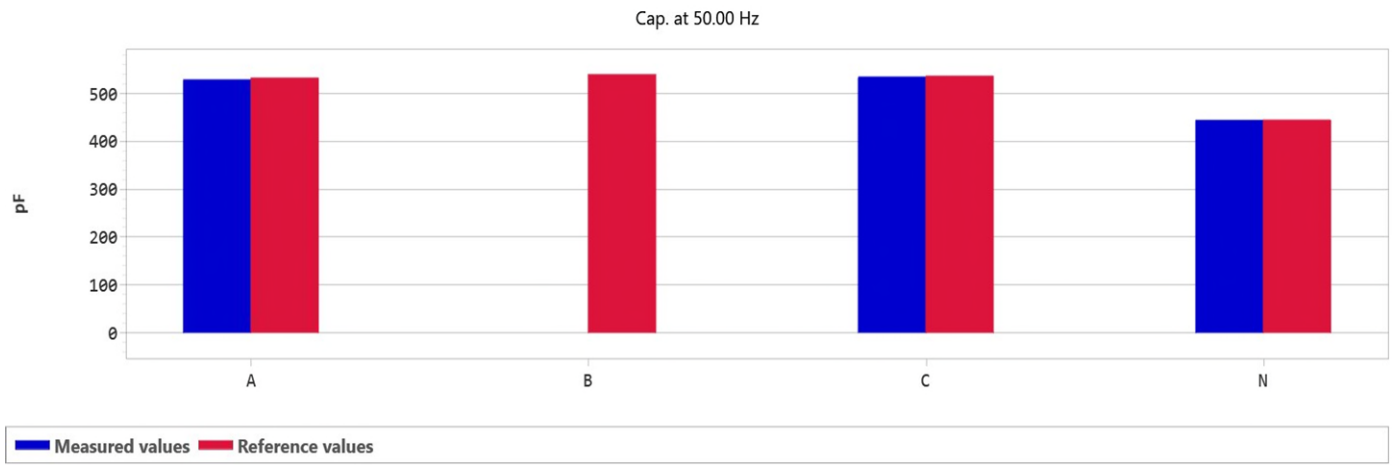
Omicron Tetrano 600 and CP TD12 unable to output on primary bushing B. Indicates an issue with this bushing.

Standard test

Corr. factor	0.8										
* Reference voltage	10.0 kV										
No.	Meas.	Test mode	Freq.	V out	* I out	* Watt losses	DF meas	DF corr	Cap. meas	Assessment	
1	A	UST-A	50.00 Hz	10.00 kV	1.66 mA	76.61 mW	0.4607 %	0.3686 %	529.3 pF	Inv.	
2	B	UST-A	50.00 Hz	kV	mA	mW	%	%	pF	Not ass.	
3	C	UST-A	50.00 Hz	10.00 kV	1.68 mA	63.21 mW	0.3760 %	0.3008 %	535.2 pF	Pass	
4	N	UST-A	50.00 Hz	10.00 kV	1.40 mA	61.10 mW	0.4376 %	0.3501 %	444.5 pF	Pass	

Graphs for standard test





Bushings Prim DF & CAP - C2			
Ambient temperature	30 °C	Weather	Clear
Top oil temperature	30 °C	Humidity	51 %

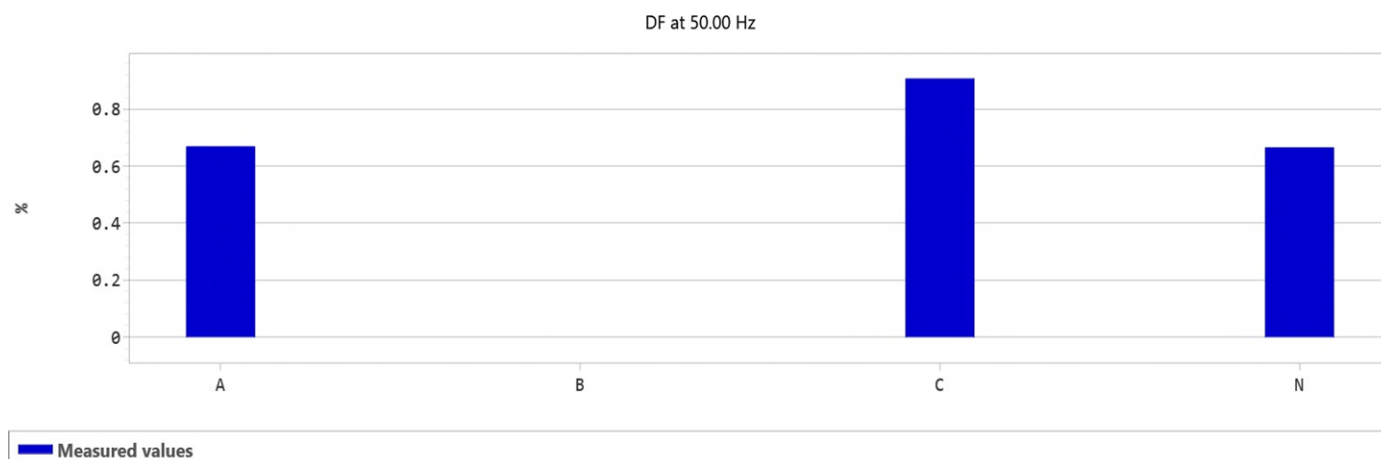
Comments

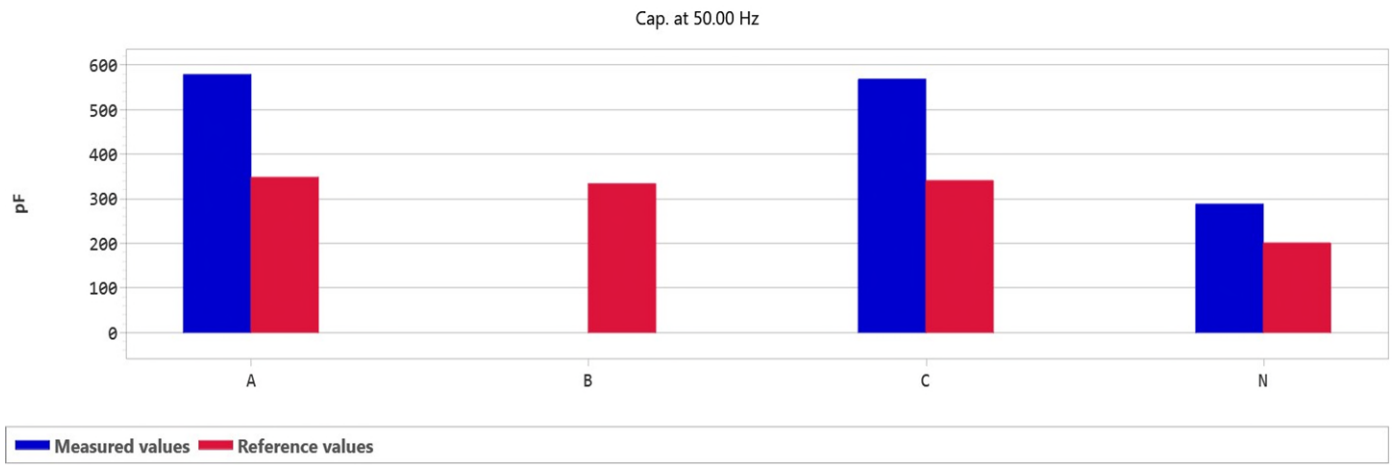
Omicron Testrano 600 and CP TD12 unable to output on primary bushing B. Indicates an issue with this bushing.

Standard test

Corr. factor		0.8									
* Reference voltage		10.0 kV									
No.	Meas.	Test mode	Freq.	V out	* I out	* Watt losses	DF meas	DF corr	Cap. meas	Assessment	
1	A	GSTg-A	50.00 Hz	0.50 kV	1.83 mA	152.68 mW	0.8356 %	0.6685 %	578.6 pF	Not ass.	
2	B	GSTg-A	50.00 Hz	kV	mA	mW	%	%	pF	Not ass.	
3	C	GSTg-A	50.00 Hz	0.50 kV	1.80 mA	203.38 mW	1.1330 %	0.9064 %	568.4 pF	Fail	
4	N	GSTg-A	50.00 Hz	0.50 kV	0.91 mA	75.91 mW	0.8305 %	0.6644 %	287.9 pF	Not ass.	

Graphs for standard test





Bushing Sec DF & CAP - C1

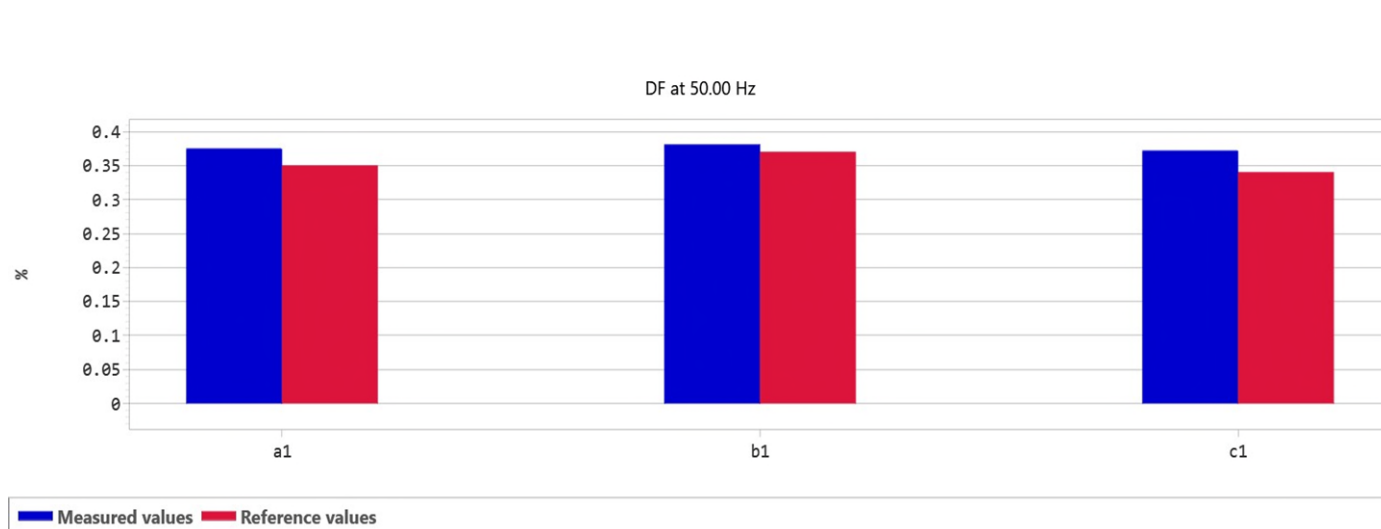
Ambient temperature	30 °C	Weather	Clear
Top oil temperature	30 °C	Humidity	51 %

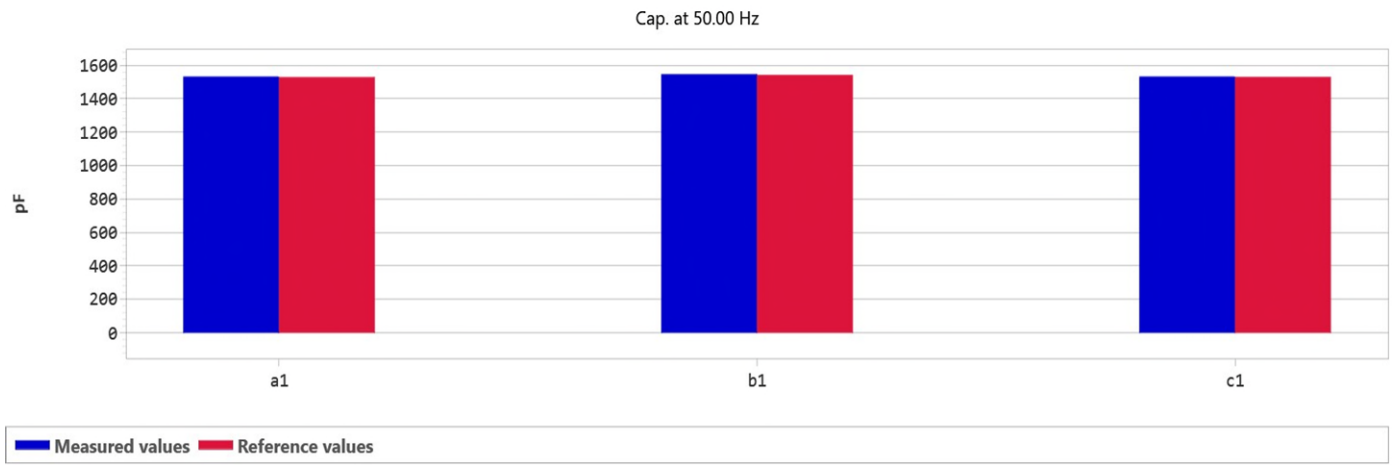
Comments

Standard test

Corr. factor		1								
* Reference voltage		10.0 kV								
No.	Meas.	Test mode	Freq.	V out	* I out	* Watt losses	DF meas	DF corr	Cap. meas	Assessment
1	a1	UST-A	50.00 Hz	10.00 kV	4.81 mA	180.33 mW	0.3749 %	0.3749 %	1531.0 pF	Pass
2	b1	UST-A	50.00 Hz	10.00 kV	4.85 mA	184.75 mW	0.3808 %	0.3808 %	1544.5 pF	Pass
3	c1	UST-A	50.00 Hz	10.00 kV	4.81 mA	178.89 mW	0.3719 %	0.3719 %	1531.2 pF	Pass

Graphs for standard test





Bushing Sec DF & CAP - C2

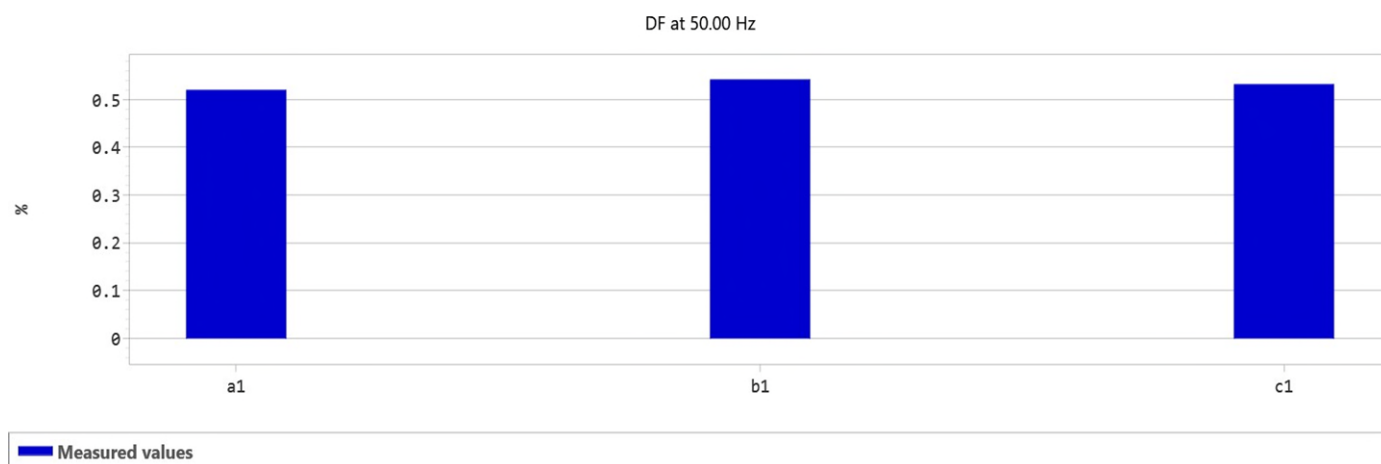
Ambient temperature	30 °C	Weather	Clear
Top oil temperature	30 °C	Humidity	51 %

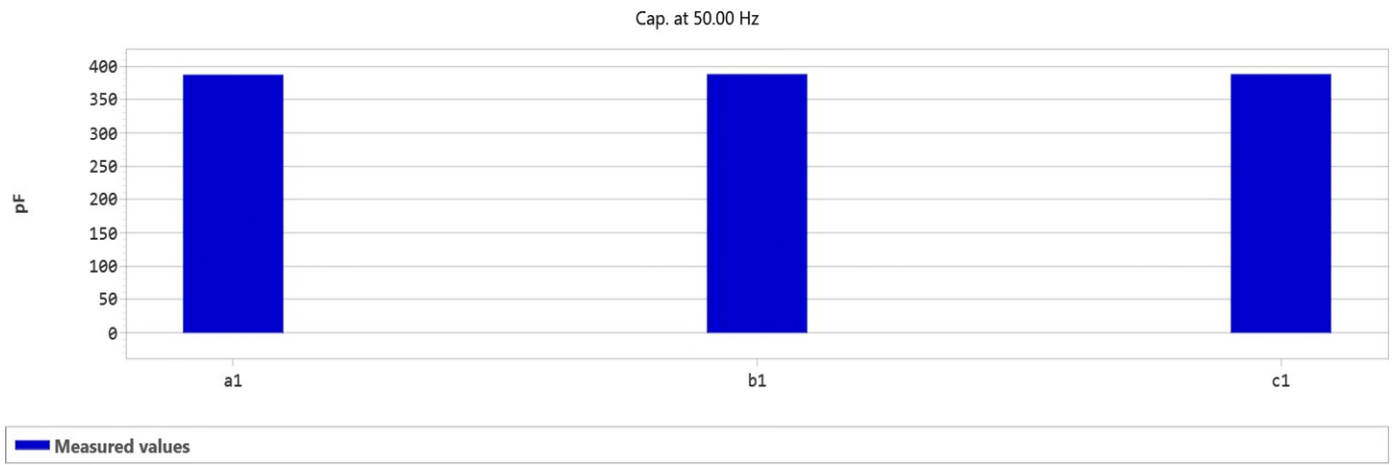
Comments

Standard test

Corr. factor		1									
* Reference voltage		10.0 kV									
No.	Meas.	Test mode	Freq.	V out	* I out	* Watt losses	DF meas	DF corr	Cap. meas	Assessment	
1	a1	GSTg-A	50.00 Hz	0.50 kV	1.22 mA	63.56 mW	0.5193 %	0.5193 %	386.6 pF	Pass	
2	b1	GSTg-A	50.00 Hz	0.50 kV	1.23 mA	66.37 mW	0.5413 %	0.5413 %	387.3 pF	Pass	
3	c1	GSTg-A	50.00 Hz	0.50 kV	1.23 mA	65.15 mW	0.5313 %	0.5313 %	387.3 pF	Pass	

Graphs for standard test





SFRA Test Tap 1			
Test voltage	10 V	Ambient temperature	27 °C
Execution date	10/10/2025		
Oil filled	Yes	Weather	Clear
Oil temperature	27 °C	Humidity	61 %
Injection	Phase	DETC position	

Test set

Serial number	HC661U	Test set version	3.0
Test set type	FRANEO 800		

Comments

Test traces

Name	Tap	Ref.	Resp.	Shorted	Grounded	Tert.	GLC1	GLC2	Time	Comment	
 1: A N	1	A	N	None	None	closed	Ω	Ω	12:40 AM		
 2: B N	1	B	N	None	None	closed	Ω	Ω	12:43 AM		
 3: C N	1	C	N	None	None	closed	Ω	Ω	12:47 AM		
 4: A N	1	A	N	a2-b2-c2	None	closed	Ω	Ω	1:02 AM		
 5: B N	1	B	N	a2-b2-c2	None	closed	Ω	Ω	12:57 AM		
 6: C N	1	C	N	a2-b2-c2	None	closed	Ω	Ω	12:52 AM		
 7: a1 b1	1 (max)	a1	b1	None	None	closed	Ω	Ω	12:16 AM		
 8: b1 c1	1	b1	c1	None	None	closed	Ω	Ω	12:21 AM		
 9: c1 a1	1	c1	a1	None	None	closed	Ω	Ω	12:23 AM		

Sweep settings

Start frequency	20.00 Hz	Receiver bandwidth	<adapt automatically>
Stop frequency	2.000 MHz	Attenuator	<adapt automatically>
Number of points	1000		
Sweep mode	Customized		

Custom frequency ranges

From frequency	To frequency	Number of points	Sweep mode
20.00 Hz	100.0 Hz	50	Linear
100.0 Hz	1.000 kHz	210	Logarithmic
1.000 kHz	10.00 kHz	210	Logarithmic
10.00 kHz	100.0 kHz	210	Logarithmic
100.0 kHz	1.000 MHz	210	Logarithmic
1.000 MHz	2.000 MHz	110	Logarithmic

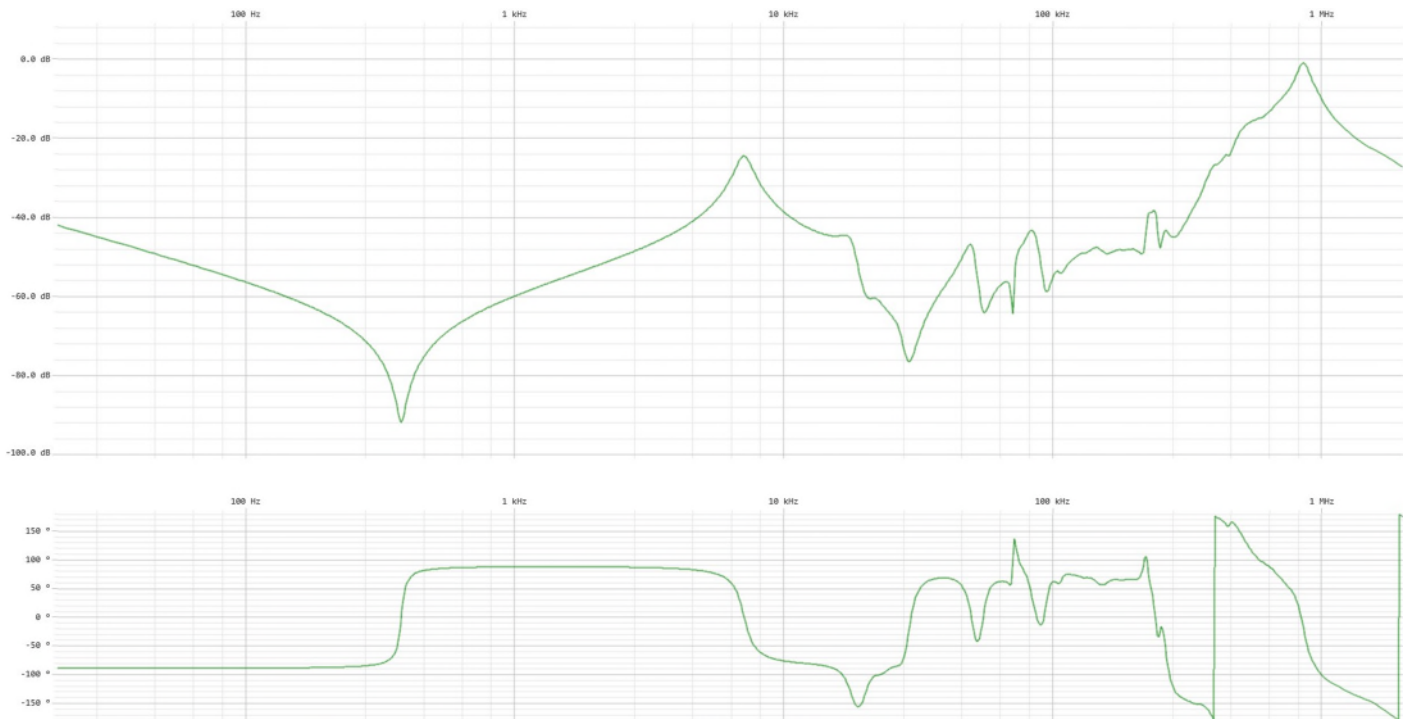
Zero Check



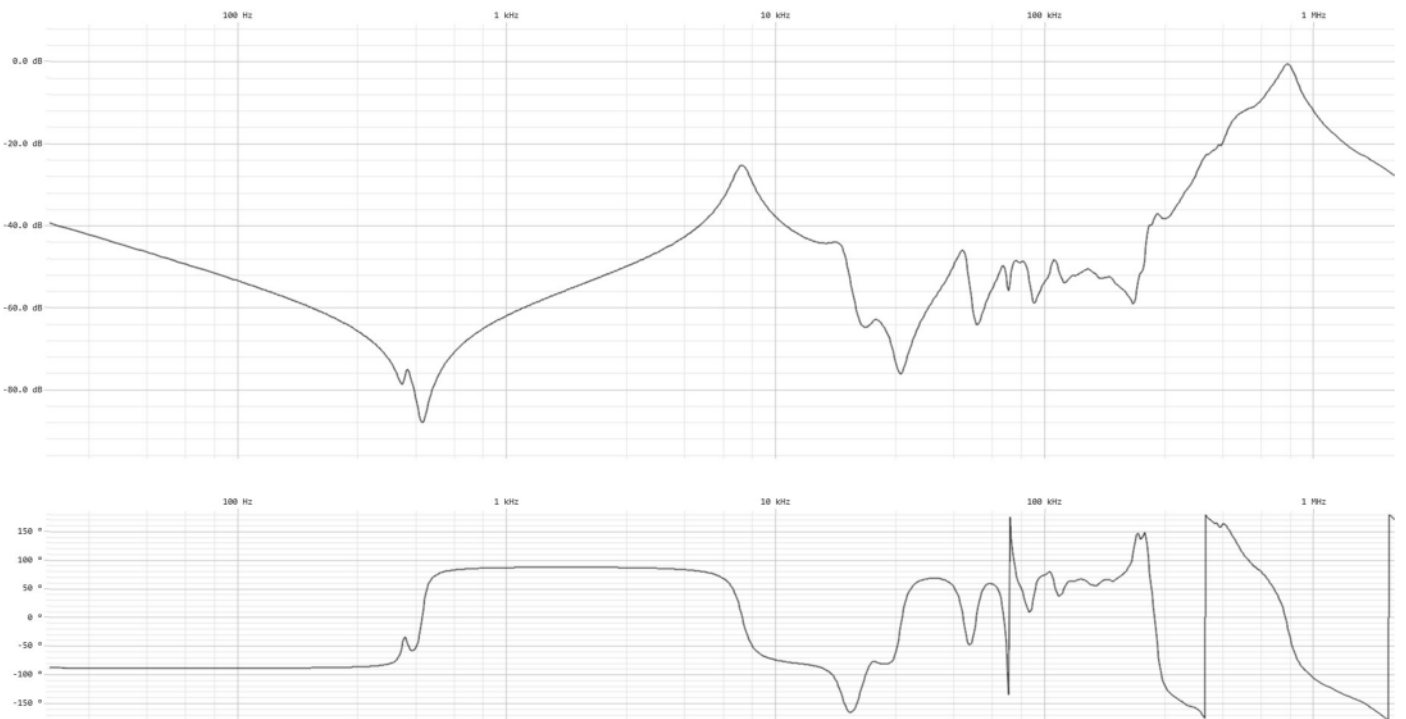
1: A N



2: B N



3: C N



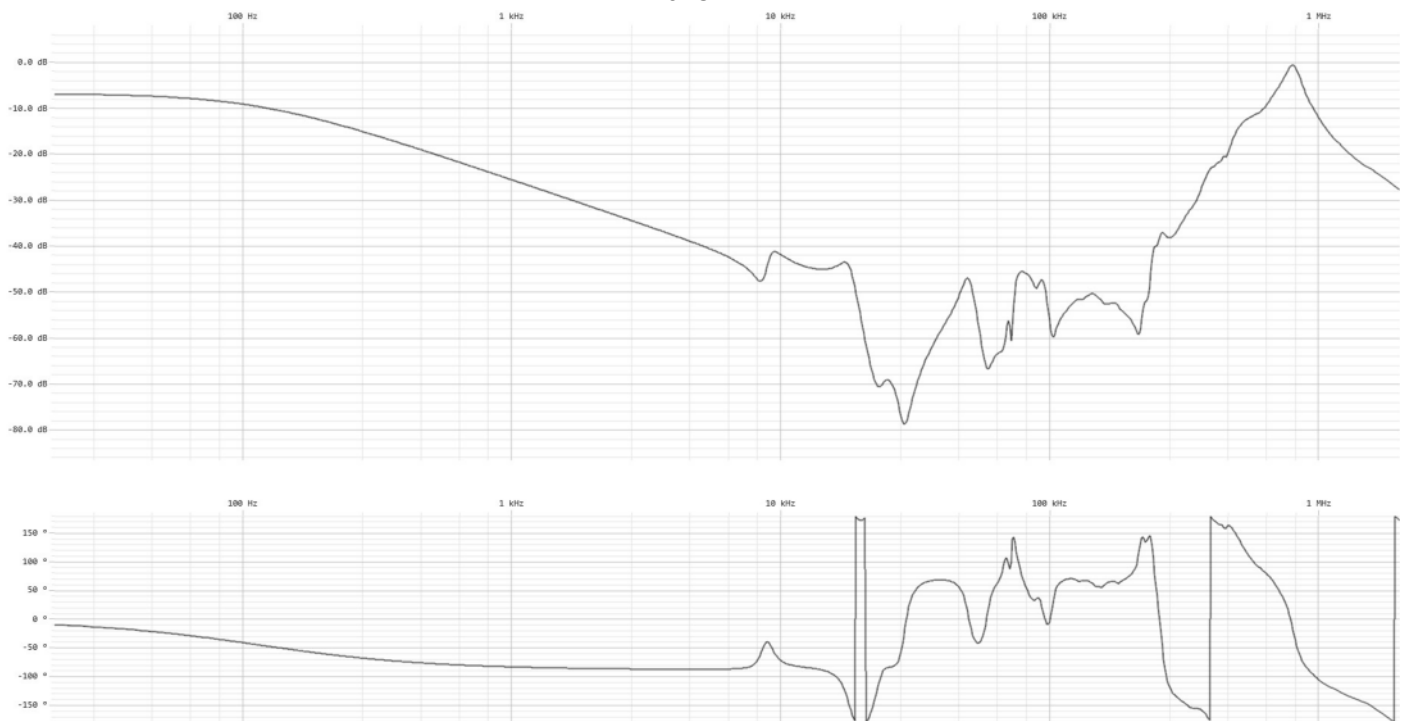
4: A N



5: B N



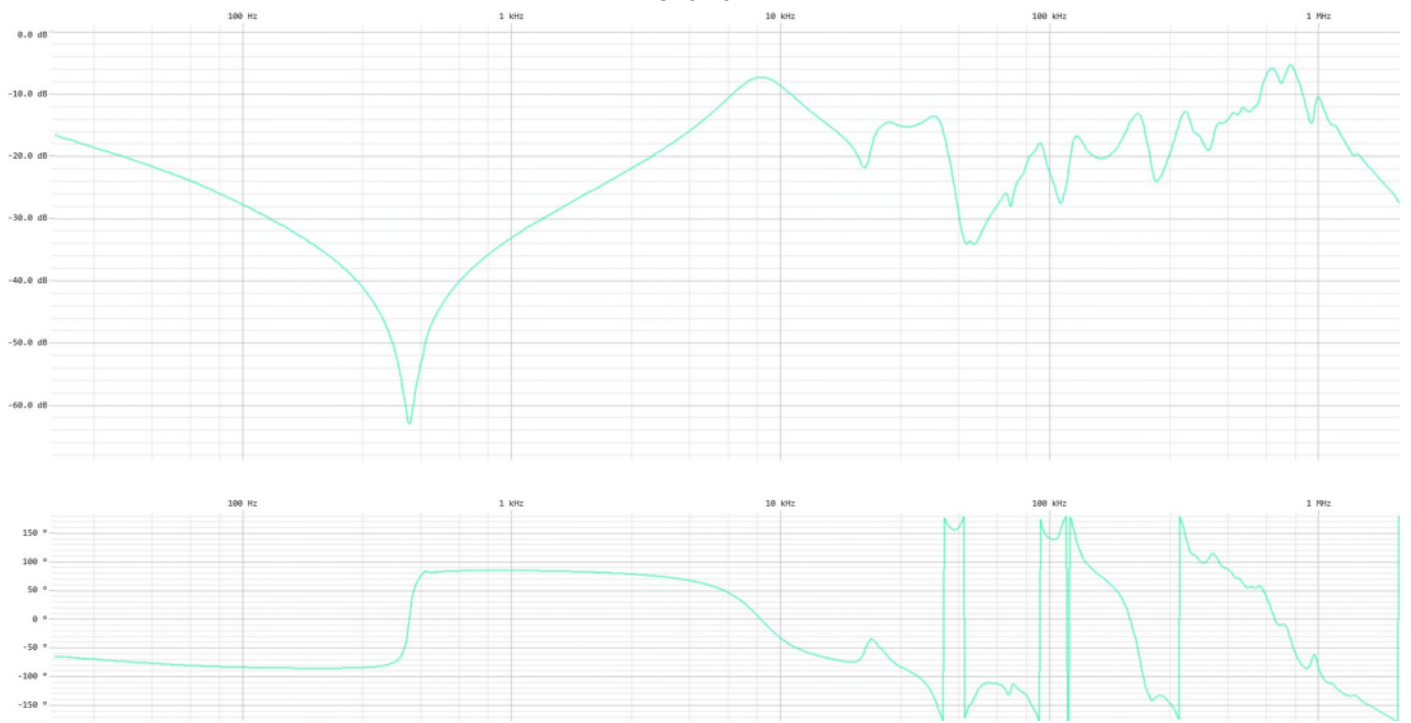
6: C N



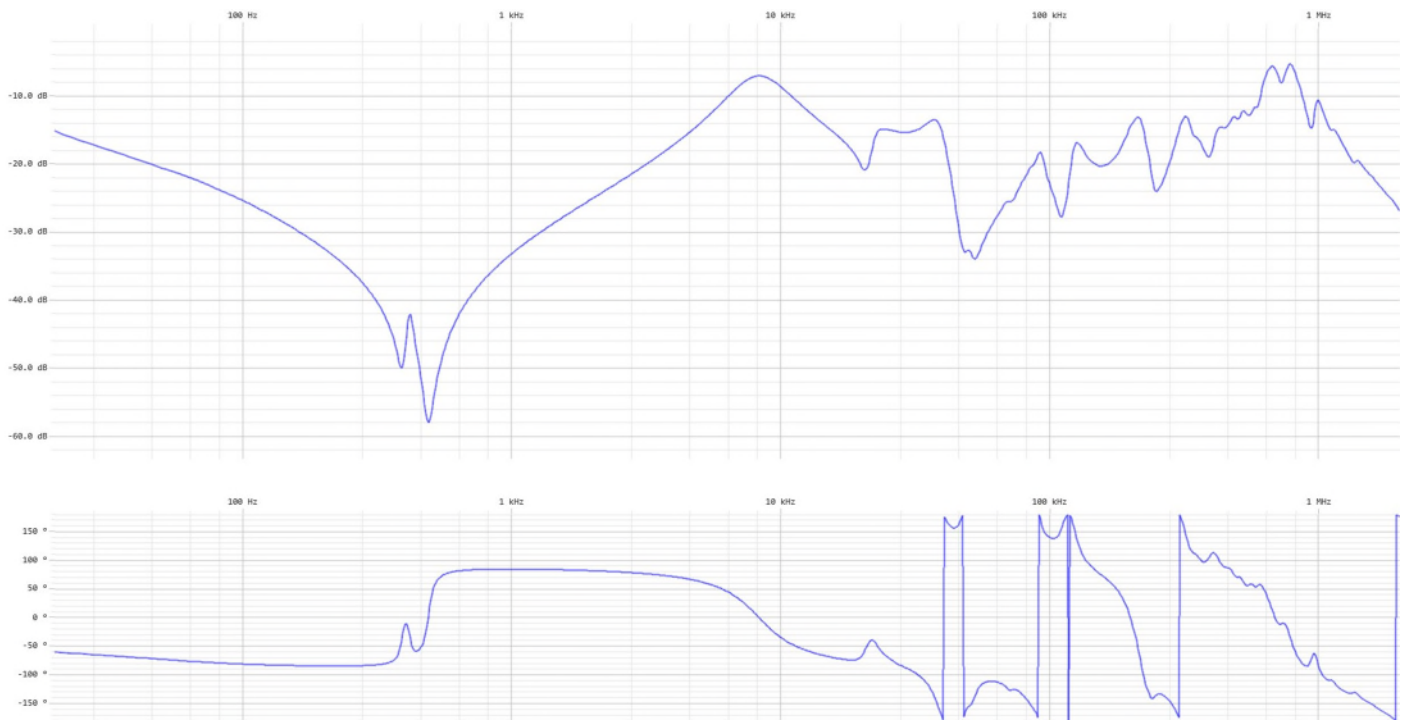
7: a1 b1



8: b1 c1



9: c1 a1



SFRA Test Tap 9

Test voltage	10 V	Ambient temperature	29 °C
Execution date	10/10/2025		
Oil filled		Weather	Clear
Oil temperature	0 °C	Humidity	54 %
Injection	Phase	DETC position	

Test set

Serial number	HC661U	Test set version	3.0
Test set type	FRANEO 800		

Comments

Test traces

Name	Tap	Ref.	Resp.	Shorted	Grounded	Tert.	GLC1	GLC2	Time	Comment	
 1: A N	9	A	N	None	None	closed	Ω	Ω	2:14 AM		
 2: B N	9	B	N	None	None	closed	Ω	Ω	2:17 AM		
 3: C N	9	C	N	None	None	closed	Ω	Ω	2:20 AM		
 4: A N	9	A	N	a2-b2-c2	None	closed	Ω	Ω	2:01 AM		
 5: B N	9	B	N	a2-b2-c2	None	closed	Ω	Ω	2:04 AM		
 6: C N	9	C	N	a2-b2-c2	None	closed	Ω	Ω	2:07 AM		
 7: a1 b1	9	a1	b1	None	None	closed	Ω	Ω	2:29 AM		
 8: b1 c1	9	b1	c1	None	None	closed	Ω	Ω	2:31 AM		
 9: c1 a1	9	c1	a1	None	None	closed	Ω	Ω	2:33 AM		

Sweep settings

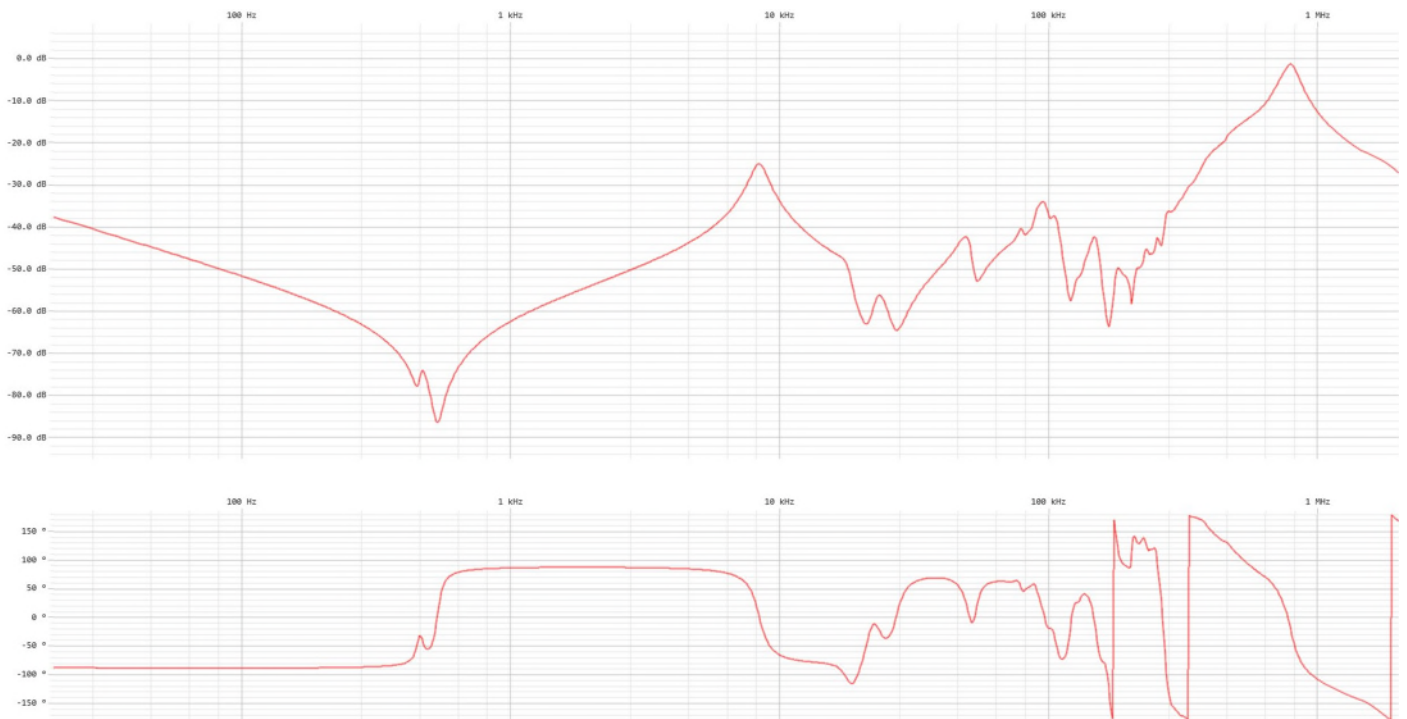
Start frequency	20.00 Hz	Receiver bandwidth	<adapt automatically>
Stop frequency	2.000 MHz	Attenuator	<adapt automatically>
Number of points	1000		
Sweep mode	Customized		

Custom frequency ranges

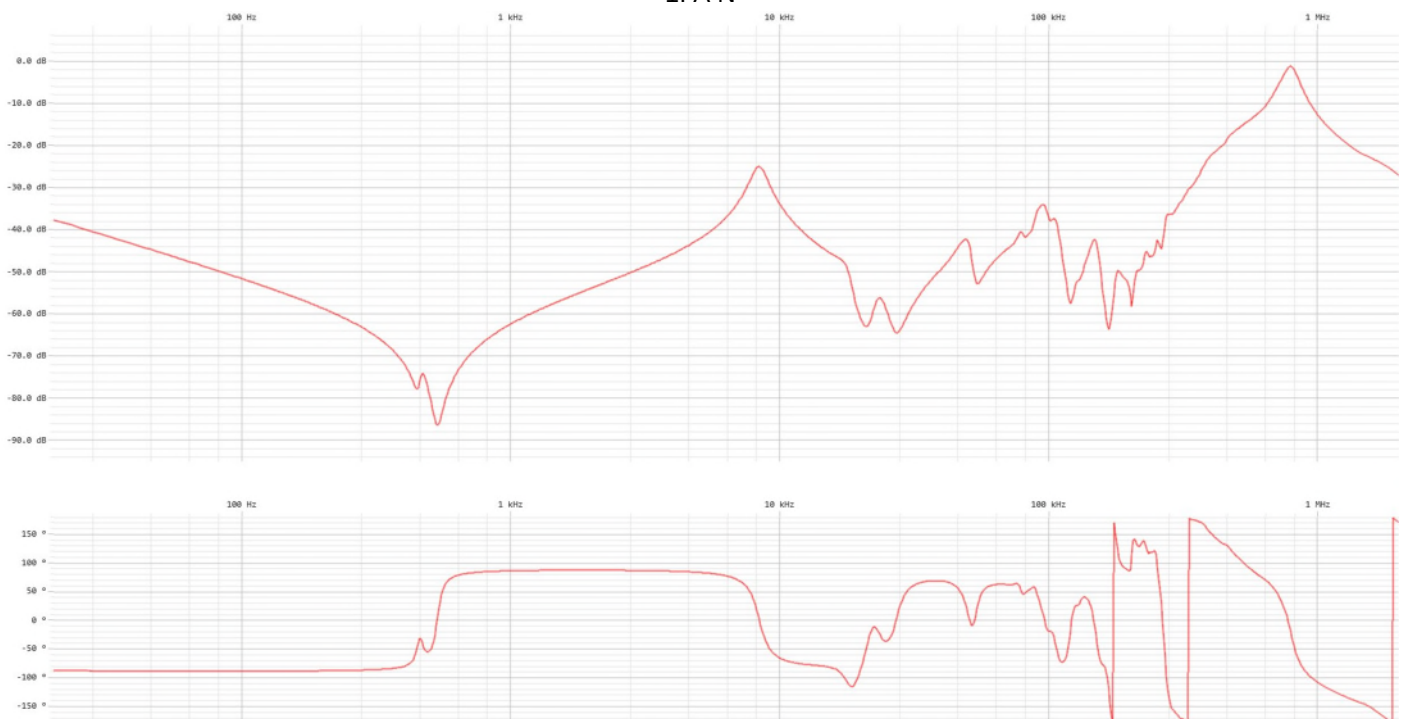
From frequency	To frequency	Number of points	Sweep mode
20.00 Hz	100.0 Hz	50	Linear
100.0 Hz	1.000 kHz	210	Logarithmic
1.000 kHz	10.00 kHz	210	Logarithmic
10.00 kHz	100.0 kHz	210	Logarithmic
100.0 kHz	1.000 MHz	210	Logarithmic
1.000 MHz	2.000 MHz	110	Logarithmic

Single traces

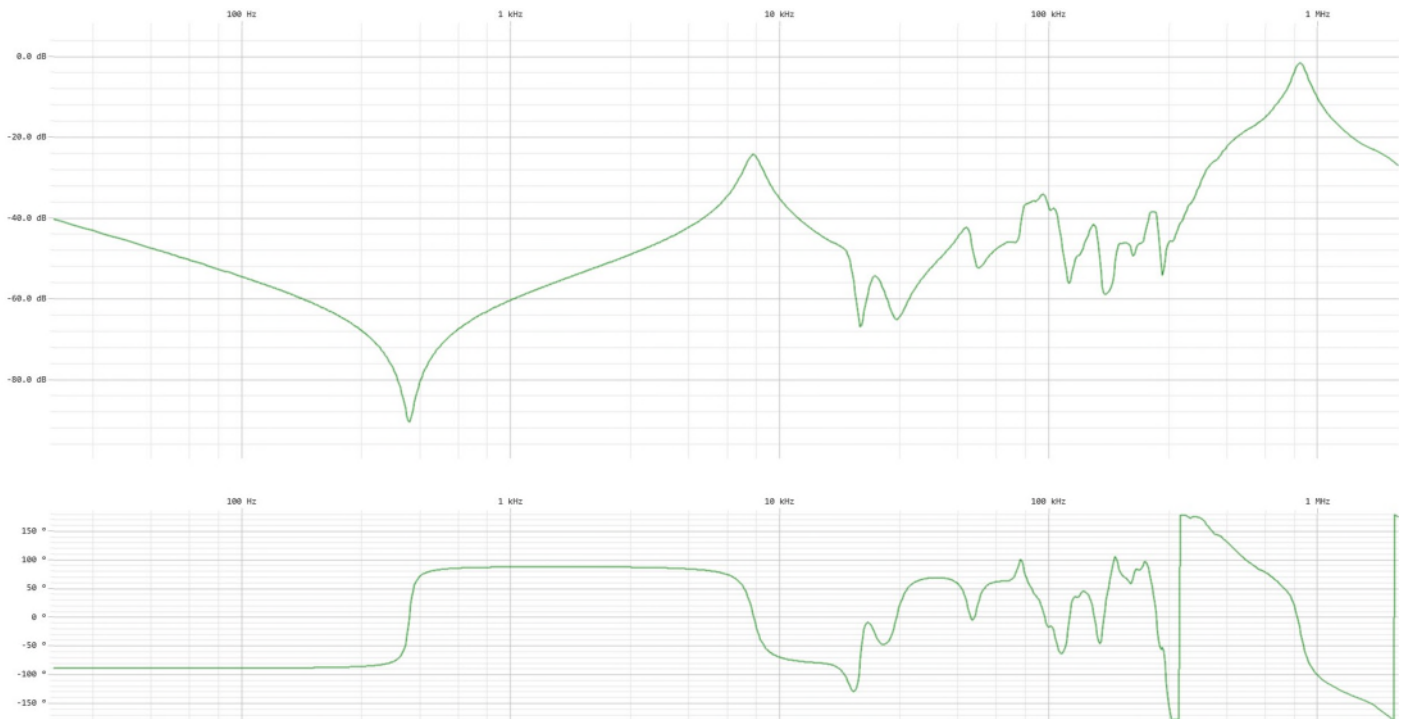
Zero Check



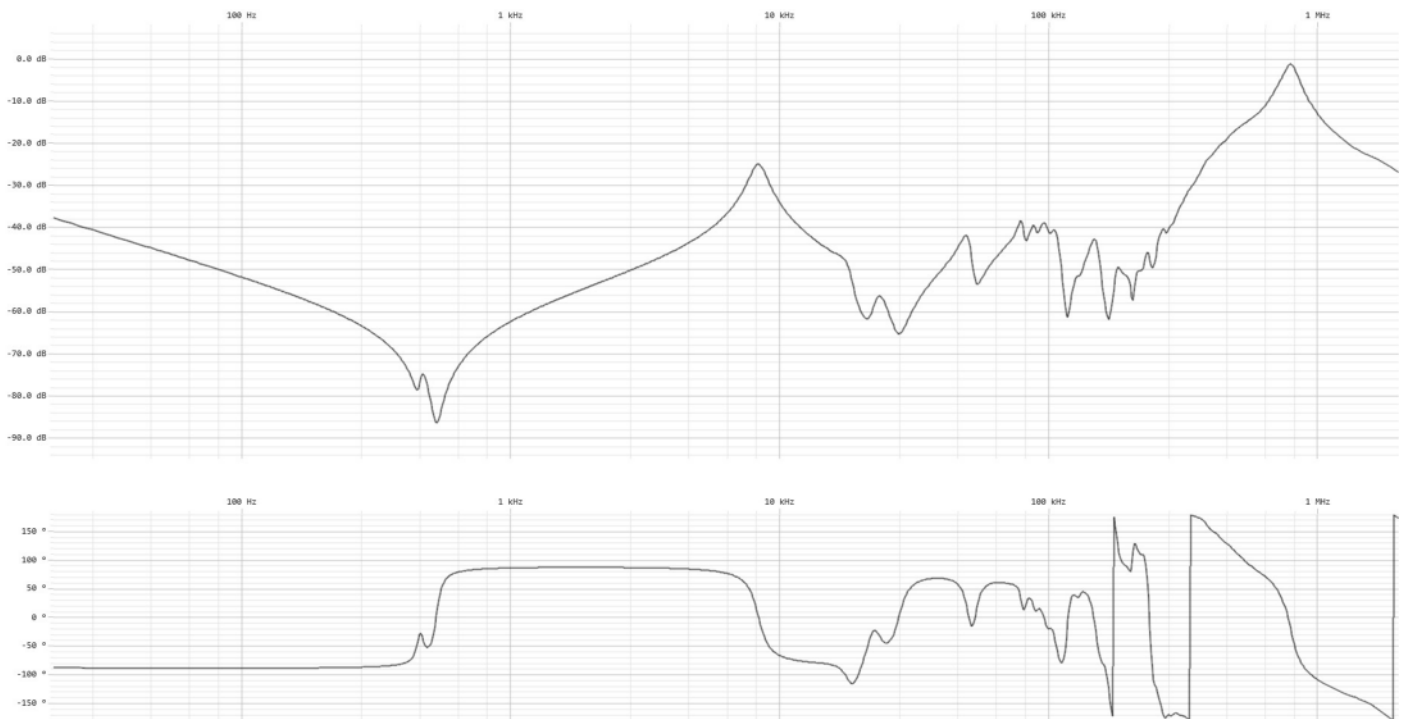
1: A N



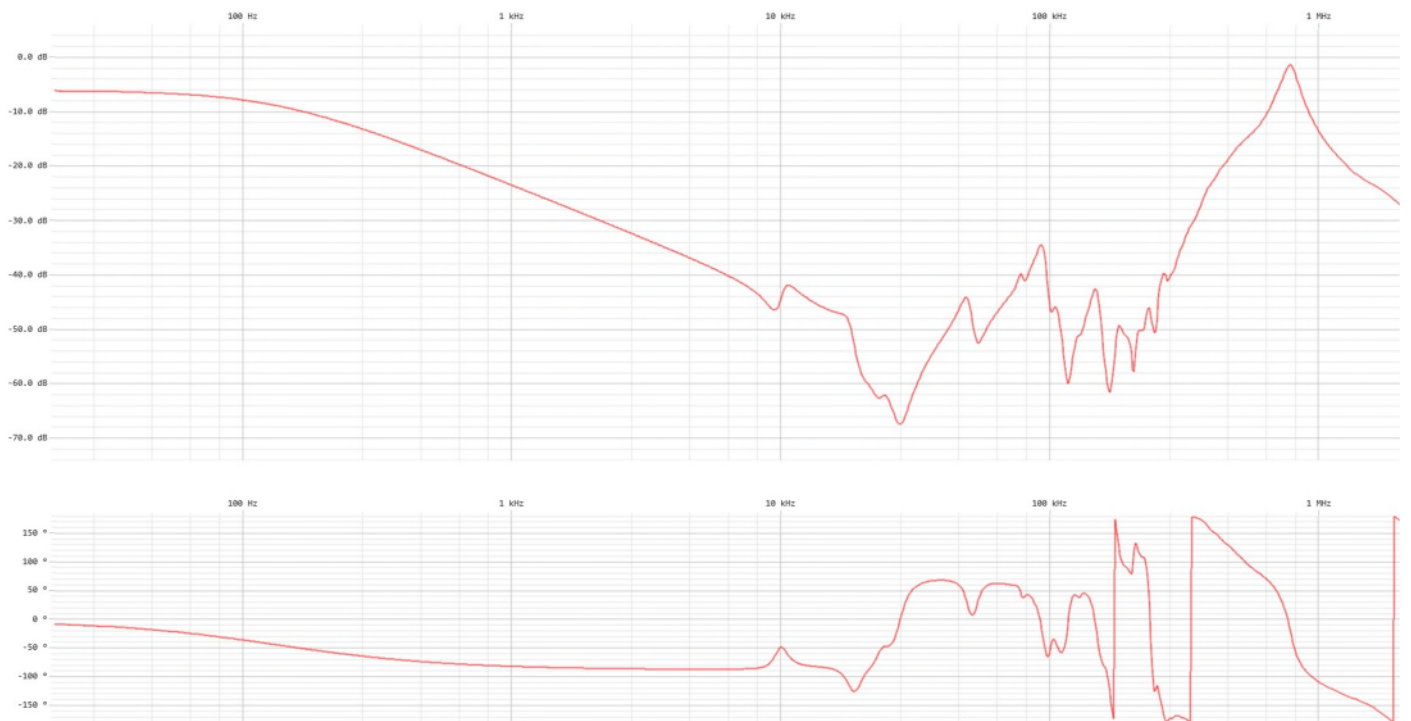
2: B N



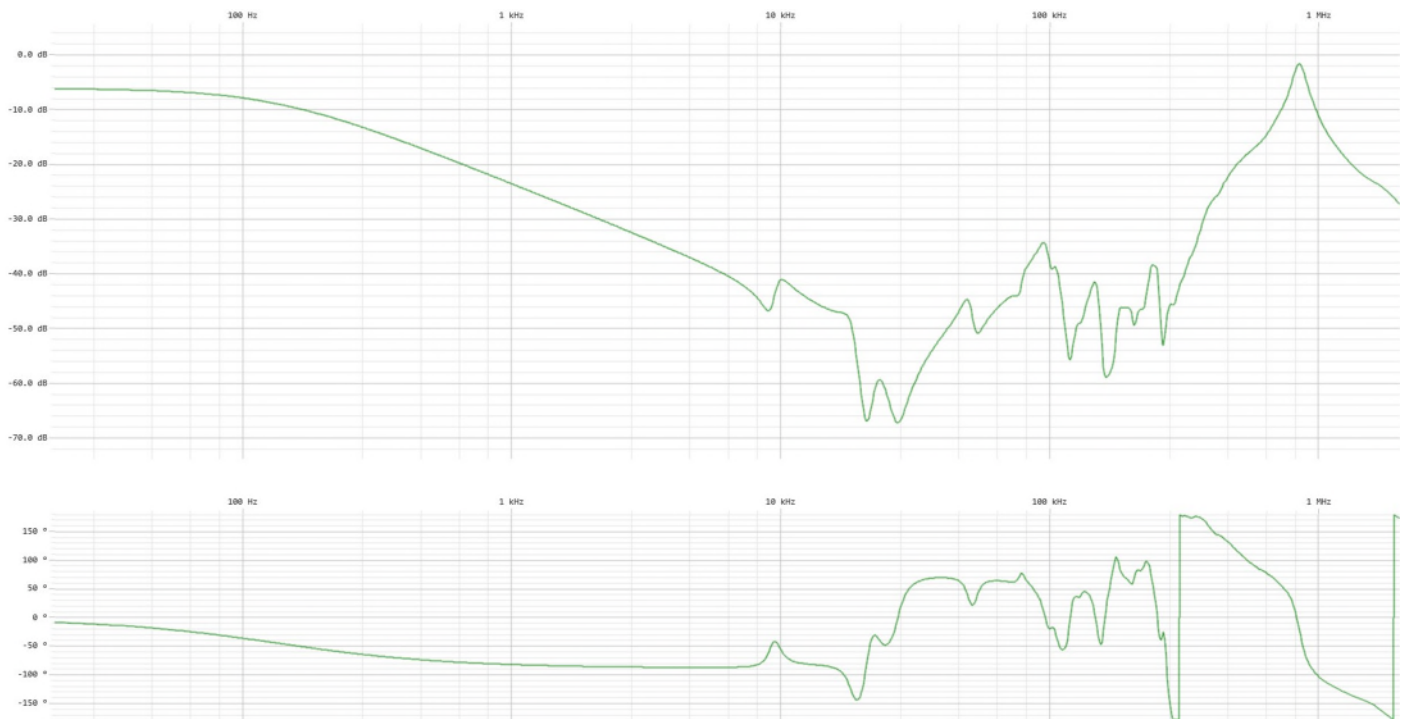
3: C N



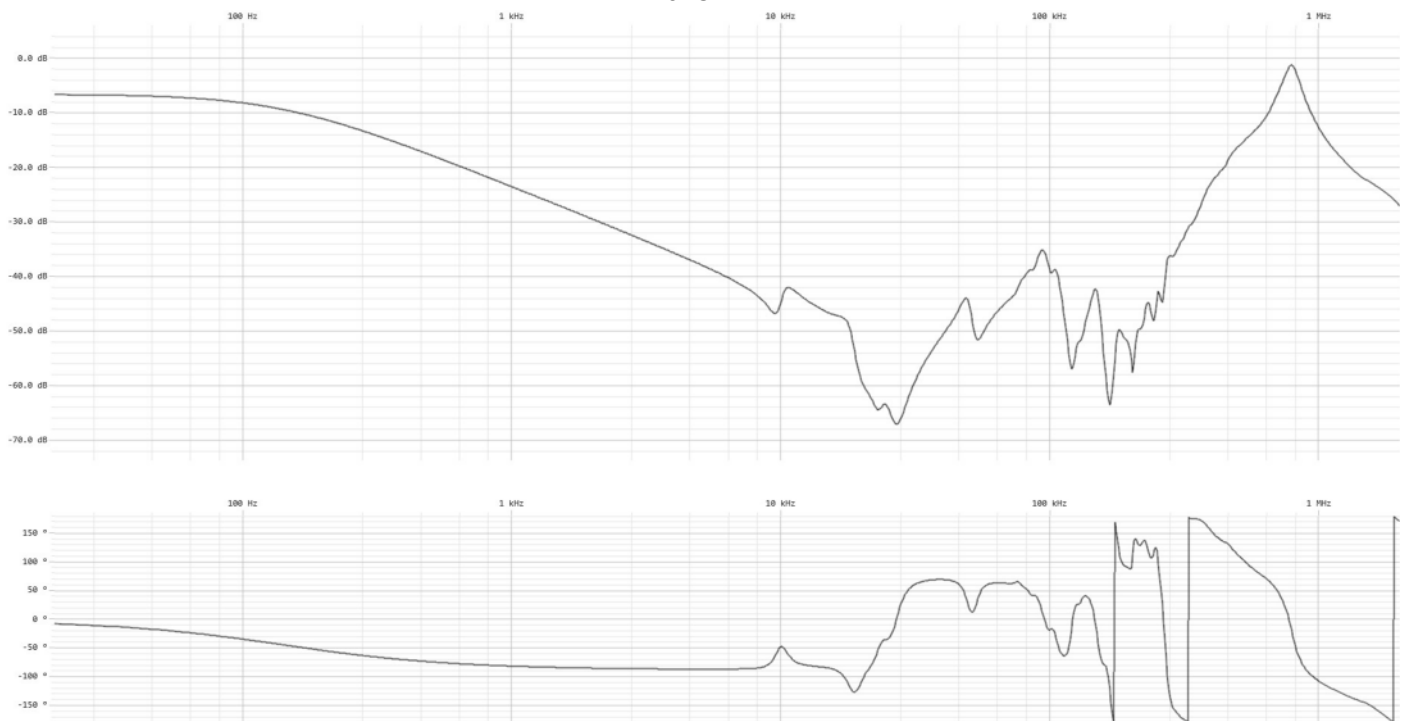
4: A N



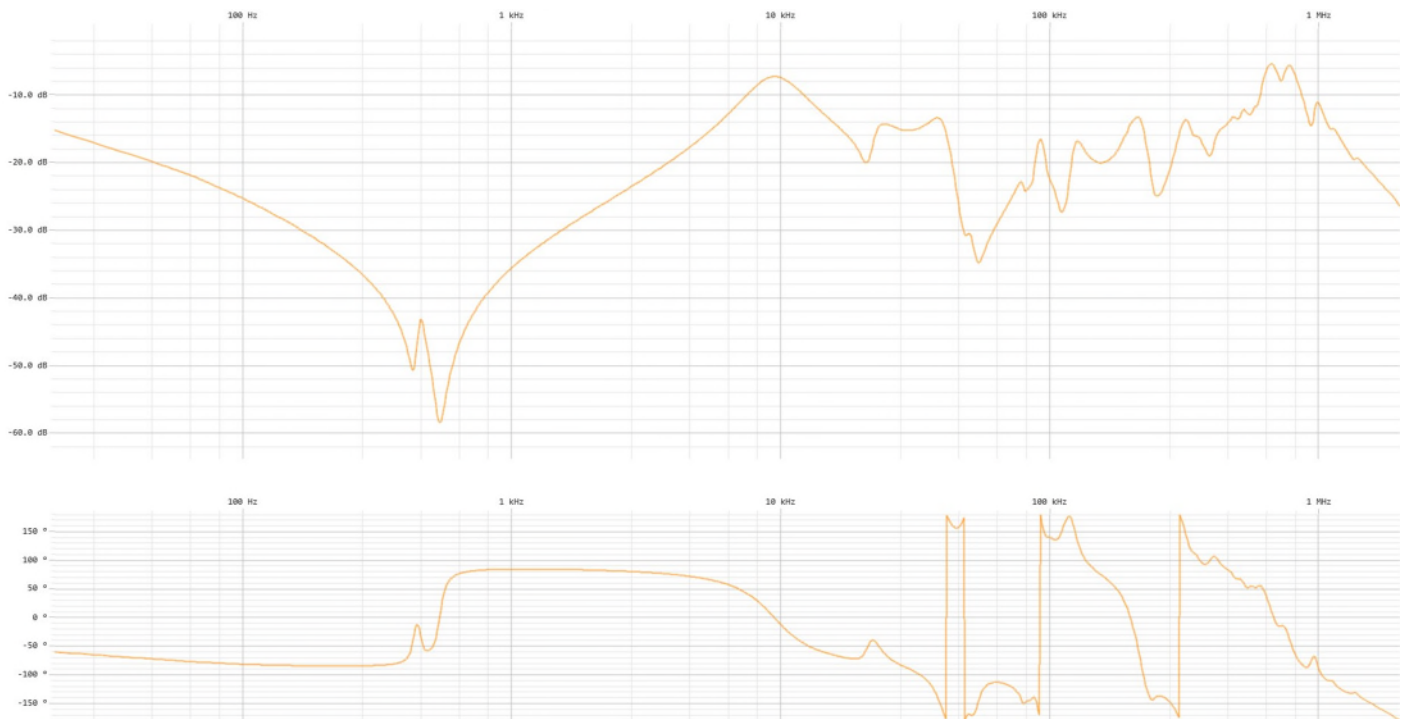
5: B N



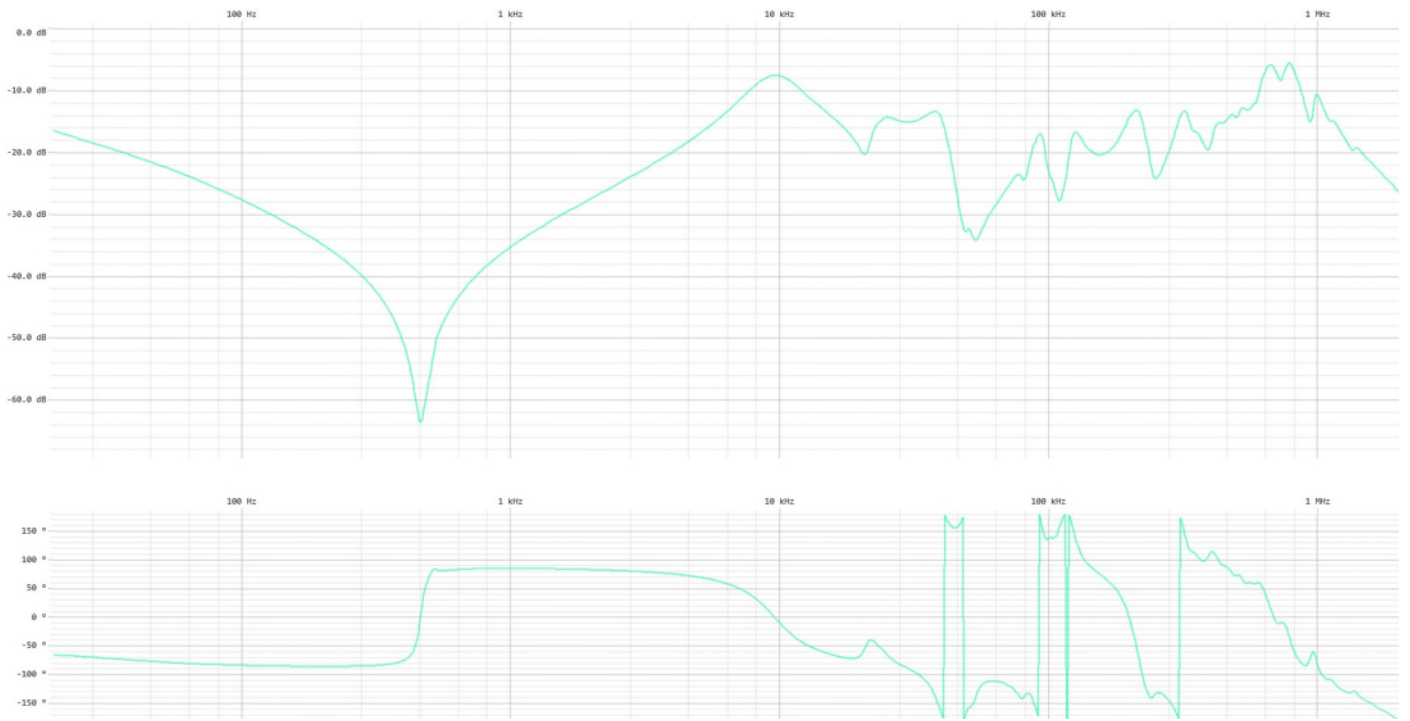
6: C N



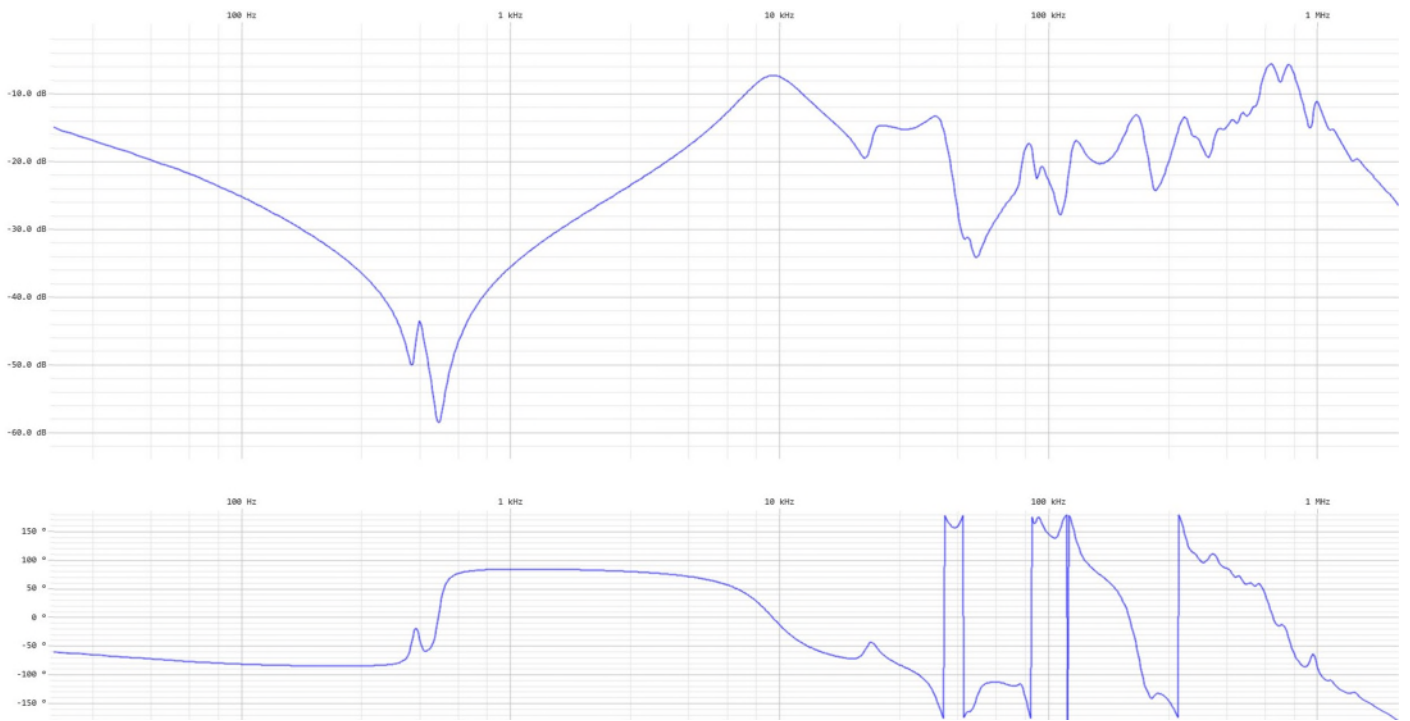
7: a1 b1



8: b1 c1



9: c1 a1



SFRA Test Tap 17

Test voltage	10 V	Ambient temperature	29 °C
Execution date	10/10/2025		
Oil filled	Yes	Weather	Clear
Oil temperature	28 °C	Humidity	55 %
Injection	Phase	DETC position	

Test set

Serial number	HC661U	Test set version	3.0
Test set type	FRANEO 800		

Comments

Test traces

Name	Tap	Ref.	Resp.	Shorted	Grounded	Tert.	GLC1	GLC2	Time	Comment	
 1: A N	17	A	N	None	None	closed	Ω	Ω	3:41 AM		
 2: B N	17	B	N	None	None	closed	Ω	Ω	3:38 AM		
 3: C N	17	C	N	None	None	closed	Ω	Ω	3:35 AM		
 4: A N	17	A	N	a2-b2-c2	None	closed	Ω	Ω	3:24 AM		
 5: B N	17	B	N	a2-b2-c2	None	closed	Ω	Ω	3:26 AM		
 6: C N	17	C	N	a2-b2-c2	None	closed	Ω	Ω	3:29 AM		
 7: a1 b1	17	a1	b1	None	None	closed	Ω	Ω	3:08 AM		
 8: b1 c1	17	b1	c1	None	None	closed	Ω	Ω	3:10 AM		
 9: c1 a1	17	c1	a1	None	None	closed	Ω	Ω	3:11 AM		

Sweep settings

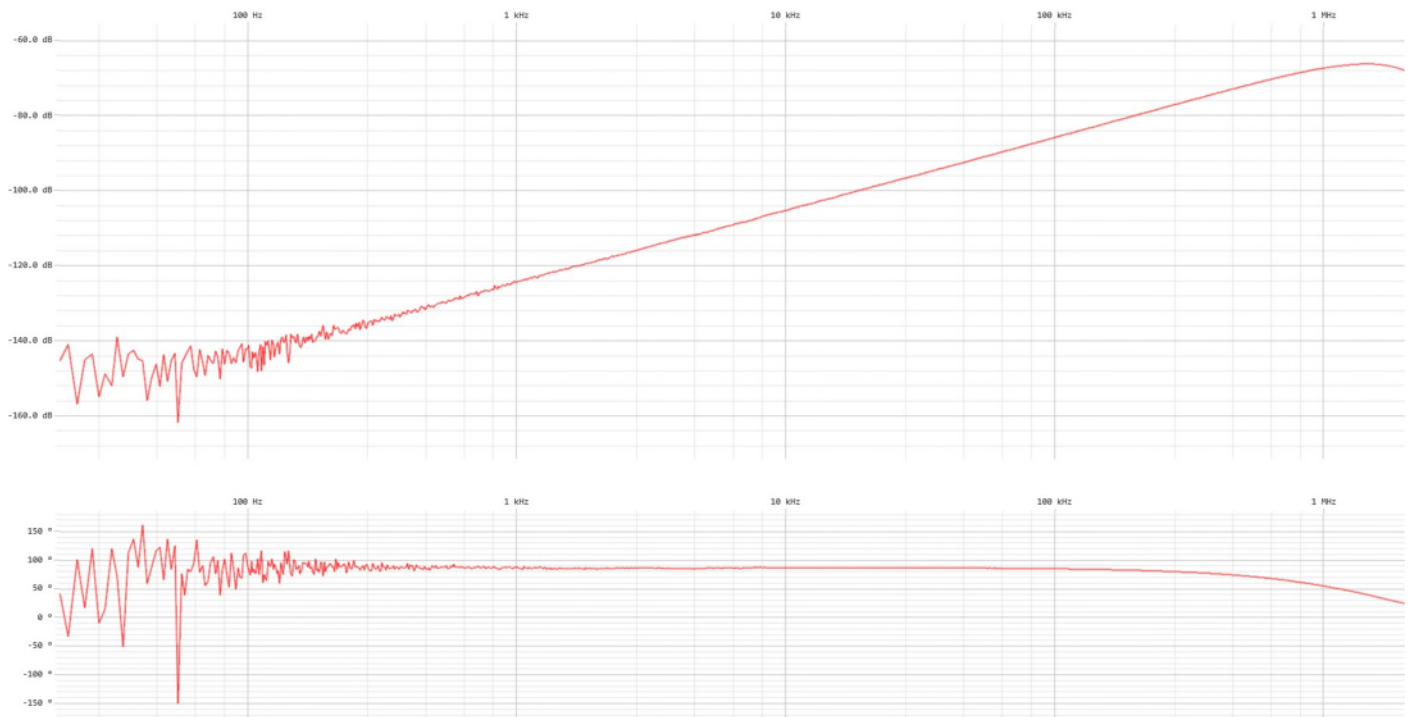
Start frequency	20.00 Hz	Receiver bandwidth	<adapt automatically>
Stop frequency	2.000 MHz	Attenuator	<adapt automatically>
Number of points	1000		
Sweep mode	Customized		

Custom frequency ranges

From frequency	To frequency	Number of points	Sweep mode
20.00 Hz	100.0 Hz	50	Linear
100.0 Hz	1.000 kHz	210	Logarithmic
1.000 kHz	10.00 kHz	210	Logarithmic
10.00 kHz	100.0 kHz	210	Logarithmic
100.0 kHz	1.000 MHz	210	Logarithmic
1.000 MHz	2.000 MHz	110	Logarithmic

Single traces

Zero Check



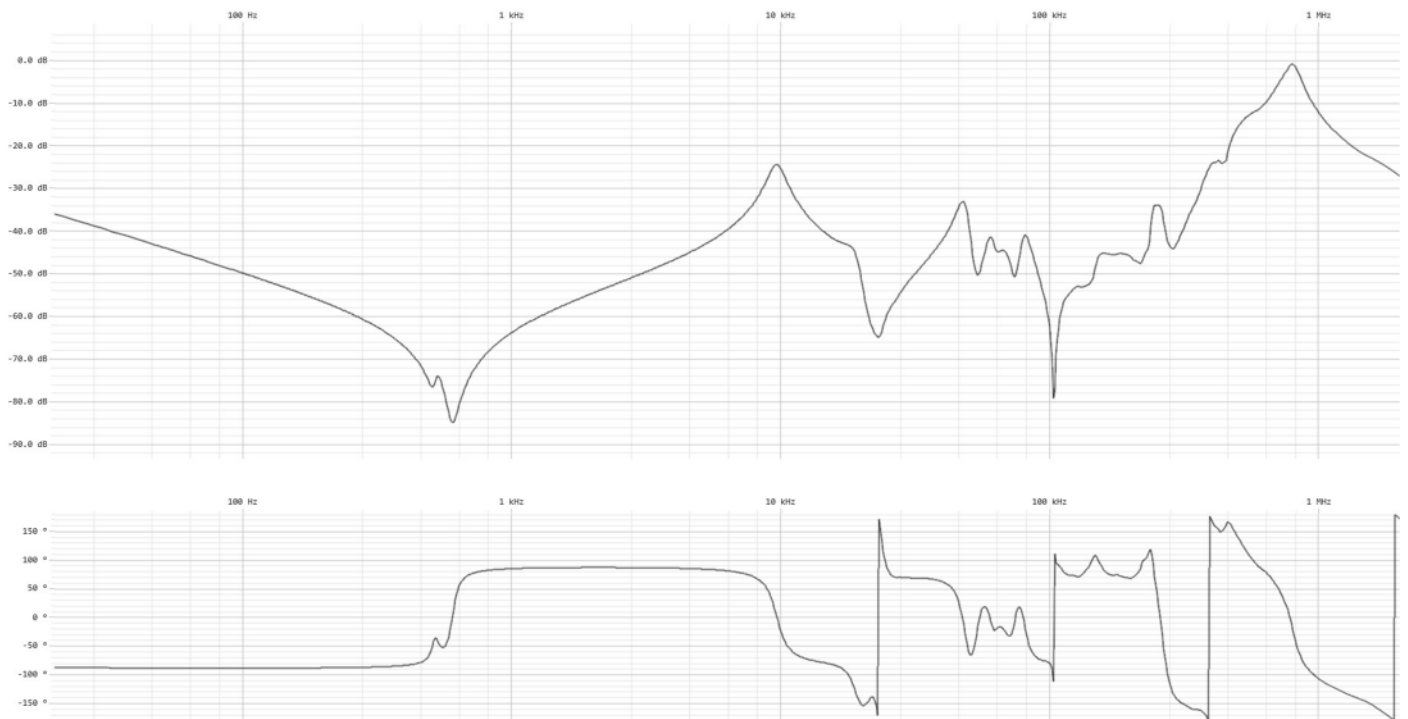
1: A N



2: B N



3: C N



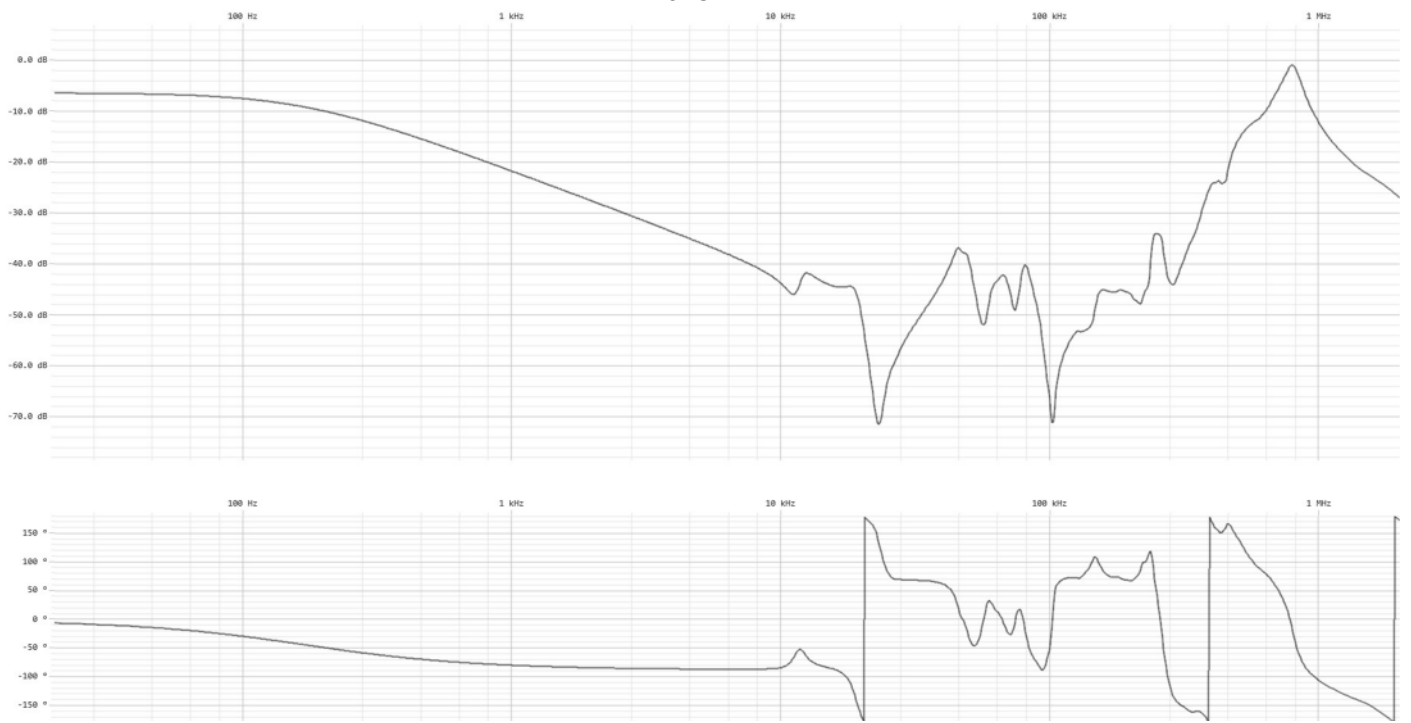
4: A N



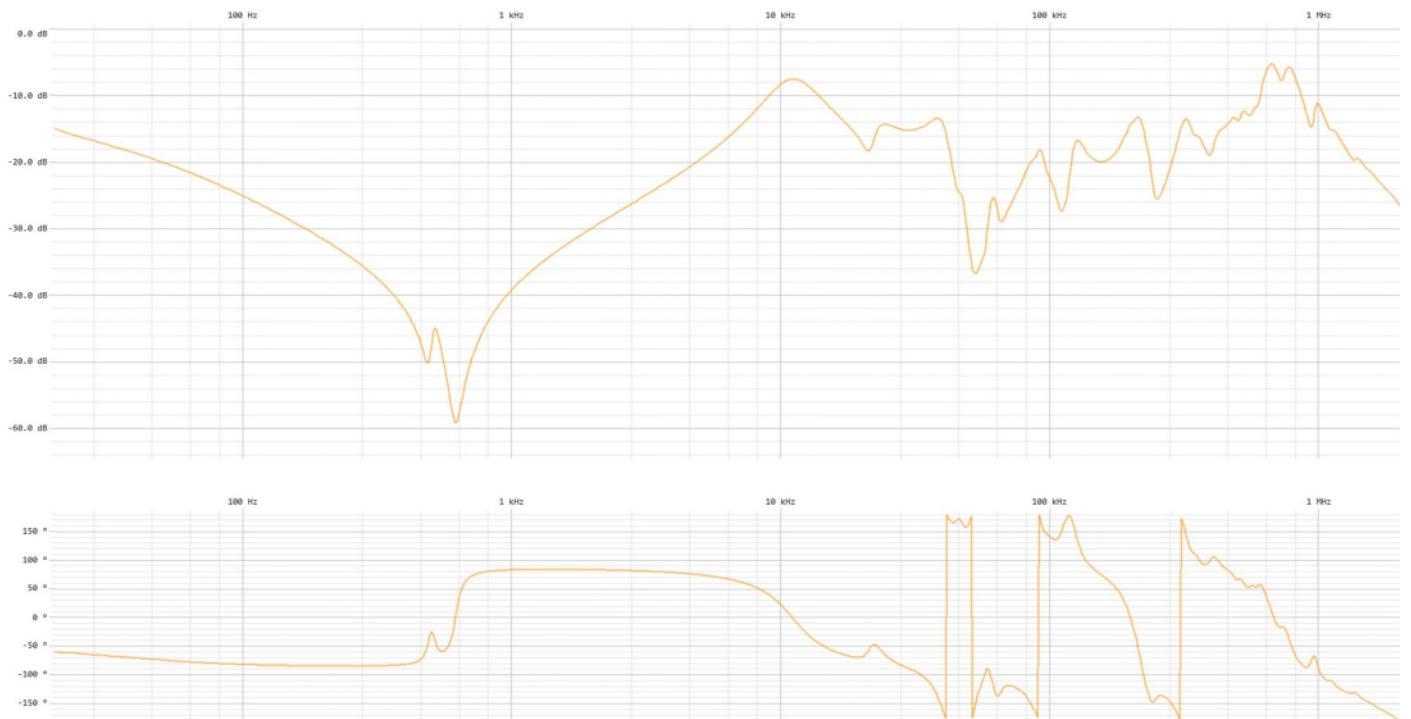
5: B N



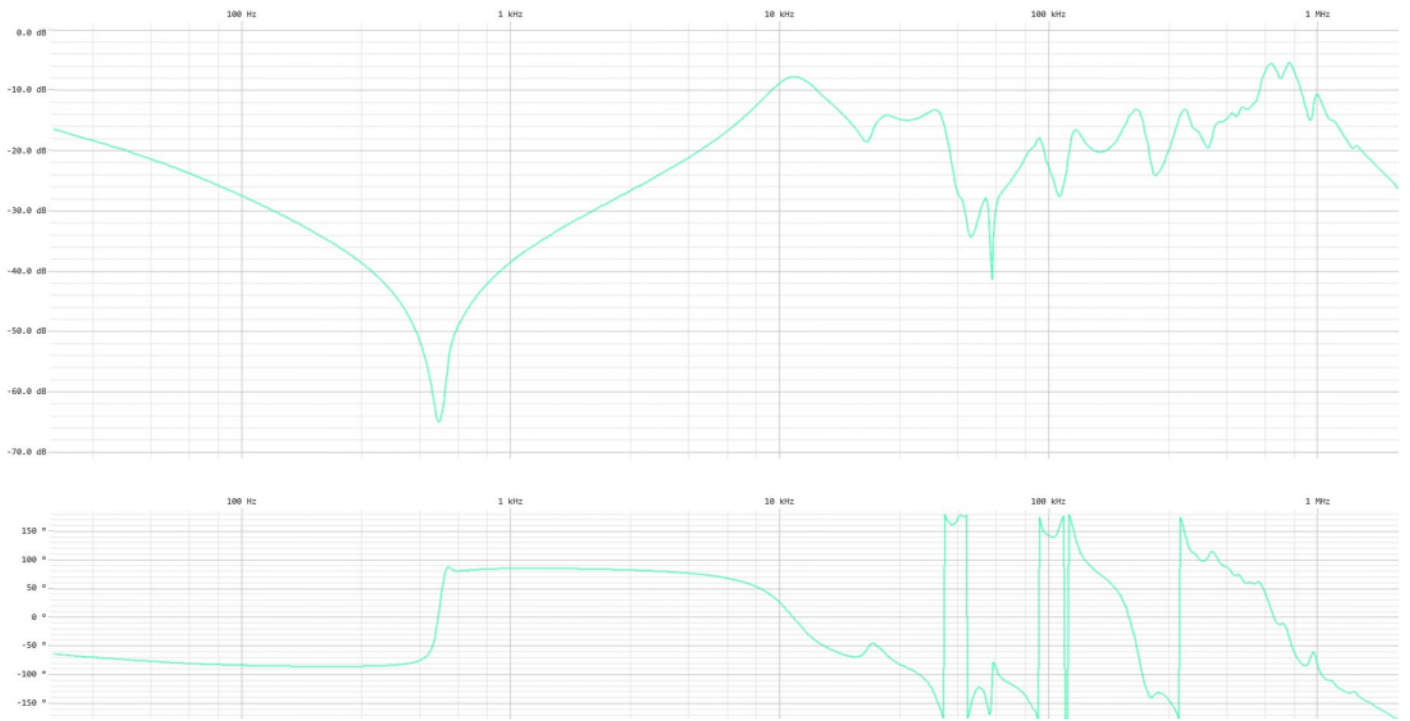
6: C N



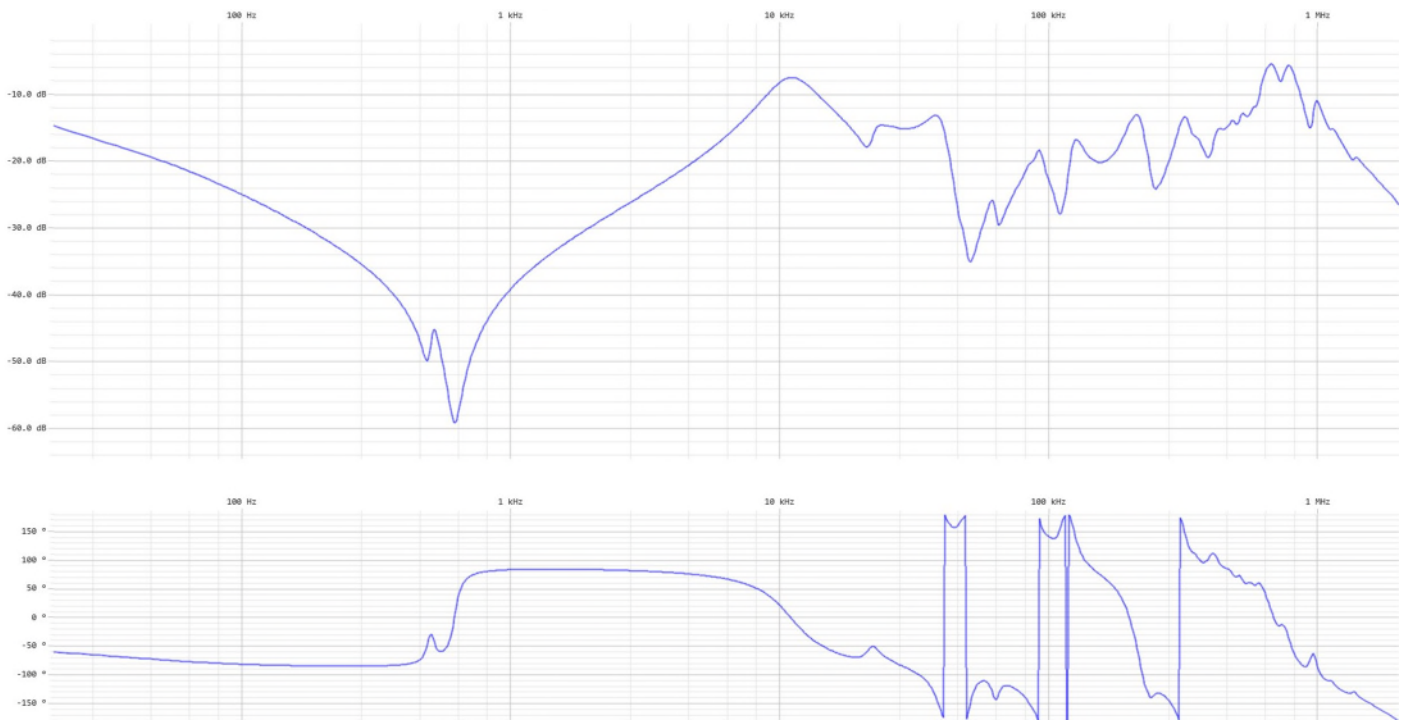
7: a1 b1



8: b1 c1



9: c1 a1



Exciting Current

Comment

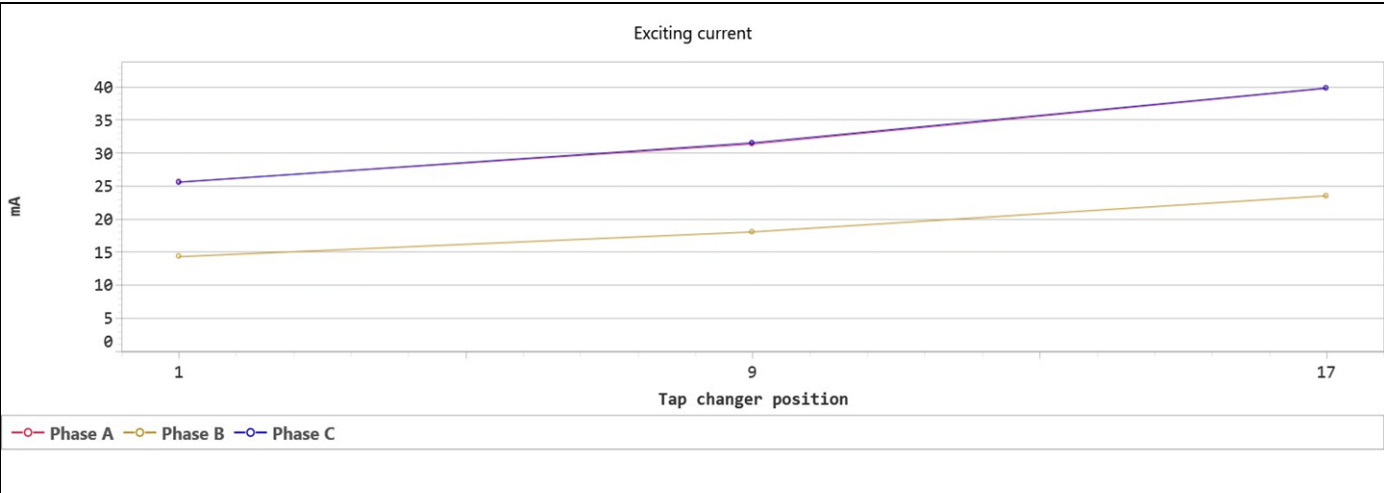
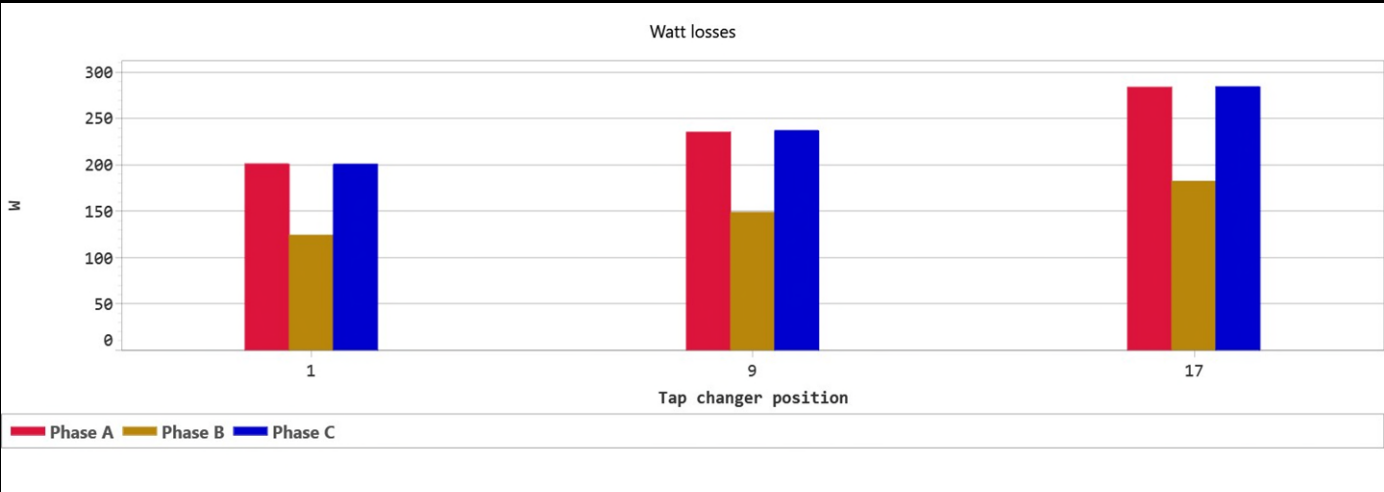
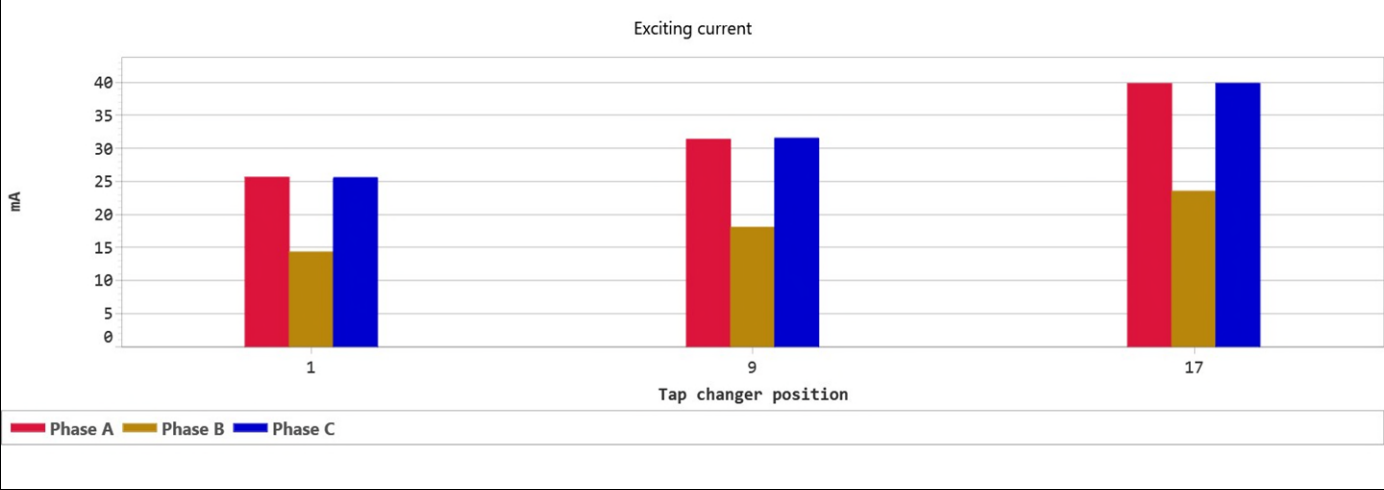
Measurement settings

Test voltage	10 kV	Automatic tap control	No
Test frequency	50.00 Hz		
Tap changer under test	OLTC		

Averaging	2	Use reference voltage	Yes
Bandwidth	±20 Hz	Reference voltage	10.00 kV
AvoidTestFrequency	No		

Measurements (sorted by tap)

Tap	Phase	V out	I out *	I phase	Watt losses *	Reactance	Assessment
1	A	10.00 kV	25.627 mA	-38.30 °	201.113 W	241.869 kΩ	Man. pass
1	B	10.04 kV	14.321 mA	-29.94 °	124.093 W	348.753 kΩ	Man. pass
1	C	10.00 kV	25.585 mA	-38.30 °	200.794 W	242.106 kΩ	Man. pass
9	A	10.00 kV	31.373 mA	-41.42 °	235.276 W	210.804 kΩ	Man. pass
9	B	9.98 kV	18.070 mA	-34.60 °	148.738 W	313.732 kΩ	Man. pass
9	C	10.00 kV	31.550 mA	-41.28 °	237.099 W	209.088 kΩ	Man. pass
17	A	10.00 kV	39.786 mA	-44.50 °	283.781 W	176.183 kΩ	Man. pass
17	B	9.96 kV	23.529 mA	-39.22 °	182.293 W	268.925 kΩ	Man. pass
17	C	10.00 kV	39.838 mA	-44.44 °	284.441 W	175.663 kΩ	Man. pass



Exciting Current Test After Demag

Comment

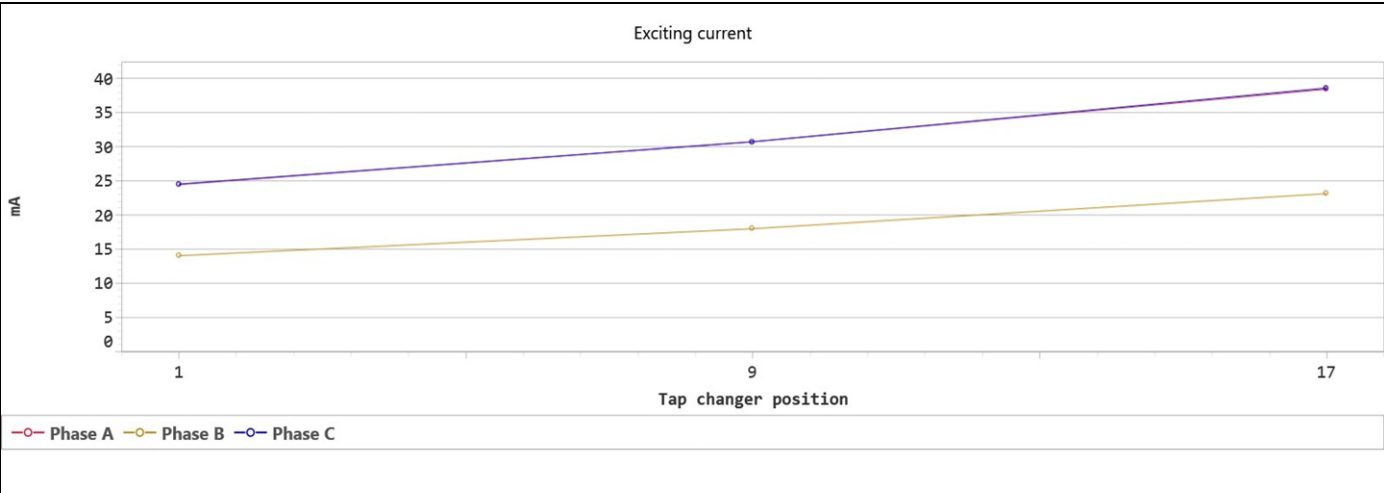
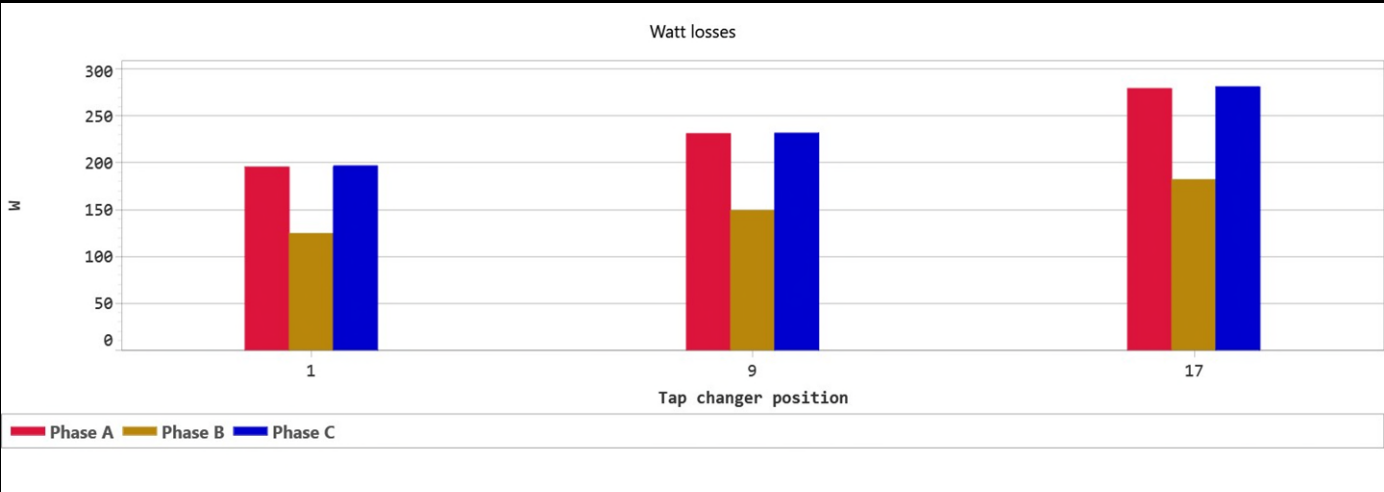
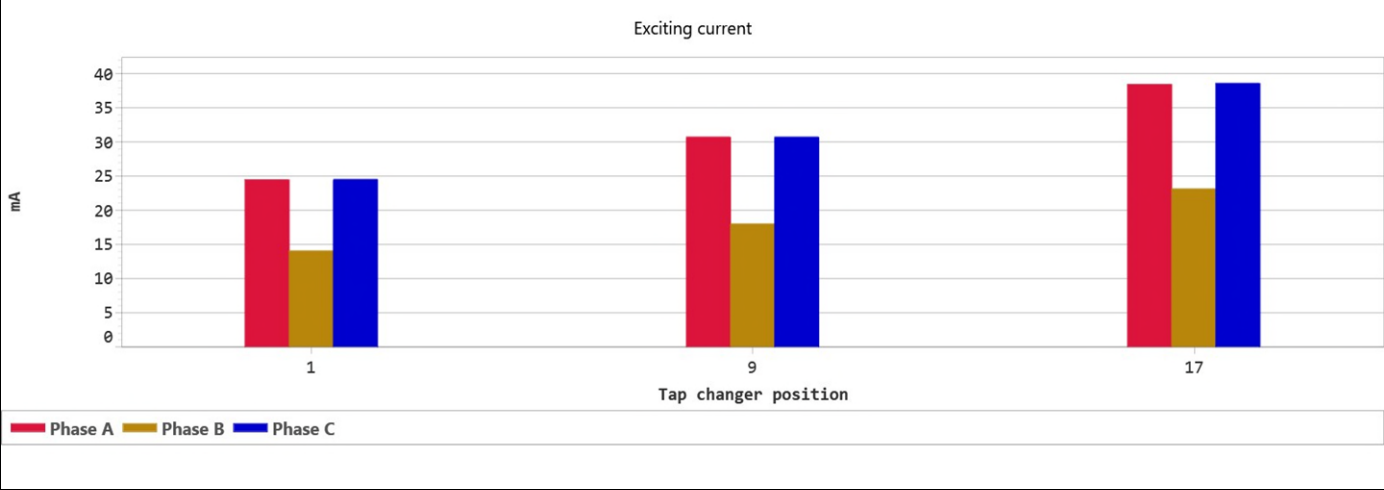
Measurement settings

Test voltage	10 kV	Automatic tap control	No
Test frequency	50.00 Hz		
Tap changer under test	OLTC		

Averaging	2	Use reference voltage	Yes
Bandwidth	±20 Hz	Reference voltage	10.00 kV
AvoidTestFrequency	No		

Measurements (sorted by tap)

Tap	Phase	V out	I out *	I phase	Watt losses *	Reactance	Assessment
1	A	10.00 kV	24.472 mA	-36.97 °	195.525 W	245.831 kΩ	Man. pass
1	B	10.03 kV	14.031 mA	-27.46 °	124.505 W	328.428 kΩ	Man. pass
1	C	10.00 kV	24.531 mA	-36.62 °	196.885 W	243.053 kΩ	Man. pass
9	A	10.00 kV	30.745 mA	-41.21 °	231.308 W	214.327 kΩ	Man. pass
9	B	10.01 kV	17.986 mA	-33.98 °	149.143 W	310.886 kΩ	Man. pass
9	C	10.00 kV	30.740 mA	-41.07 °	231.745 W	213.714 kΩ	Man. pass
17	A	10.00 kV	38.455 mA	-43.38 °	279.489 W	178.578 kΩ	Man. pass
17	B	9.99 kV	23.129 mA	-38.10 °	182.002 W	266.582 kΩ	Man. pass
17	C	10.00 kV	38.623 mA	-43.24 °	281.363 W	177.423 kΩ	Man. pass



Ratio Prim-Sec

Comment

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Measurement settings

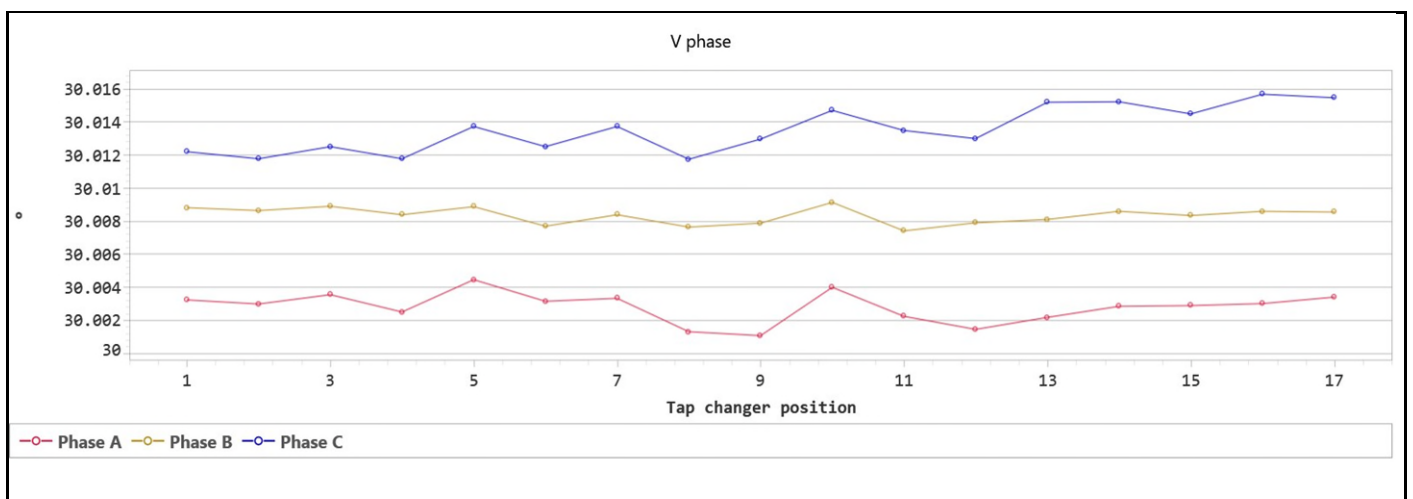
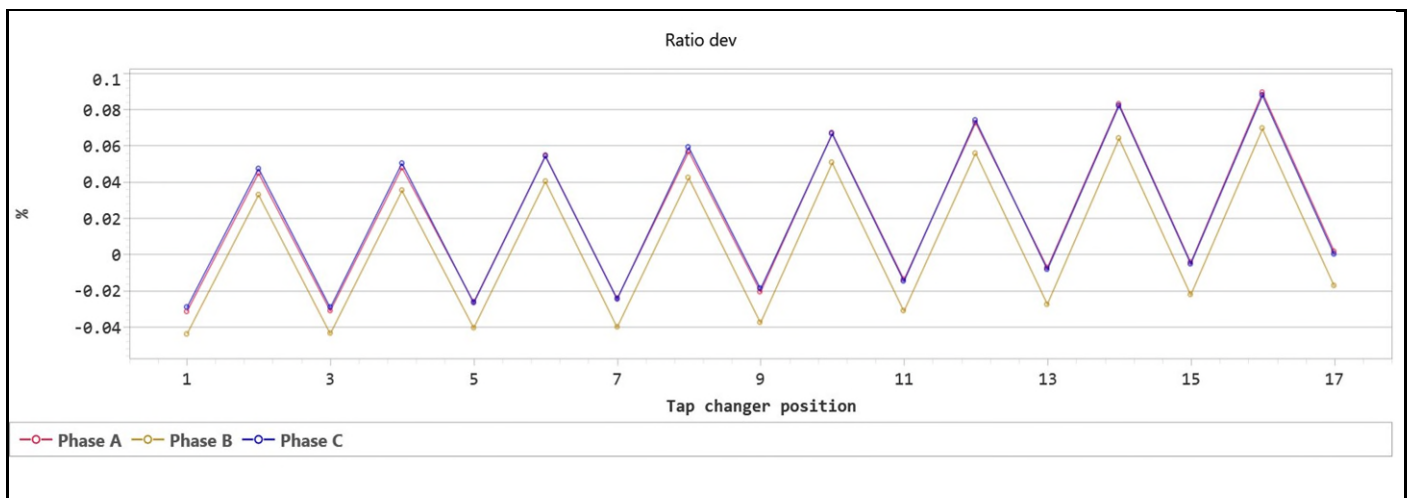
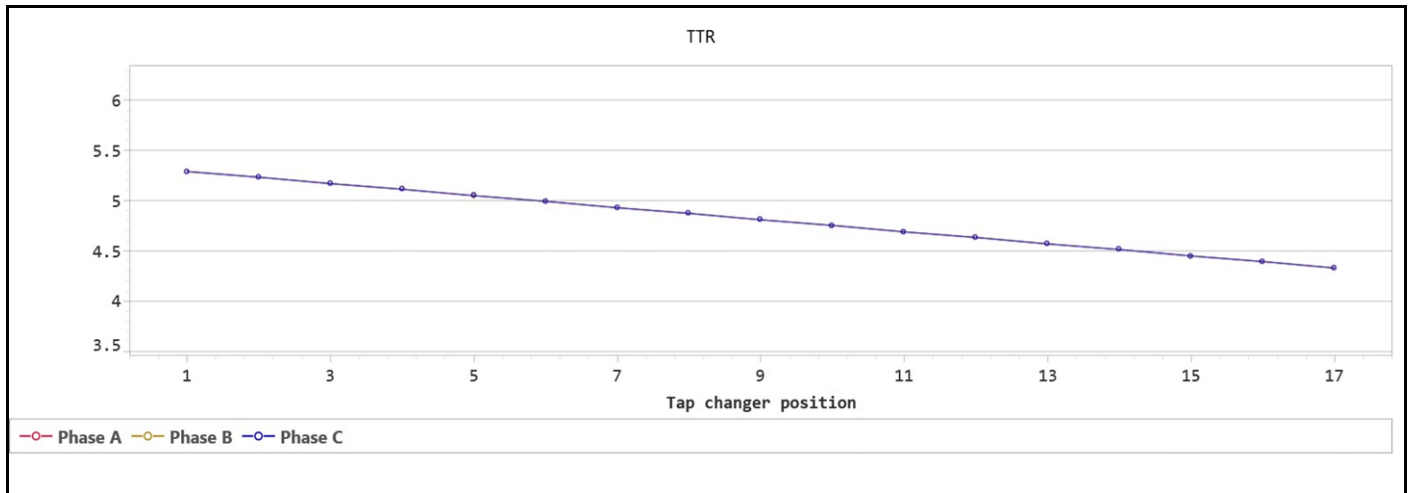
Test voltage	120 V	Automatic tap control	No
Test frequency	50.00 Hz		
Tap changer under test	OLTC		
		Ratio	TTR*

Measurements (sorted by tap)

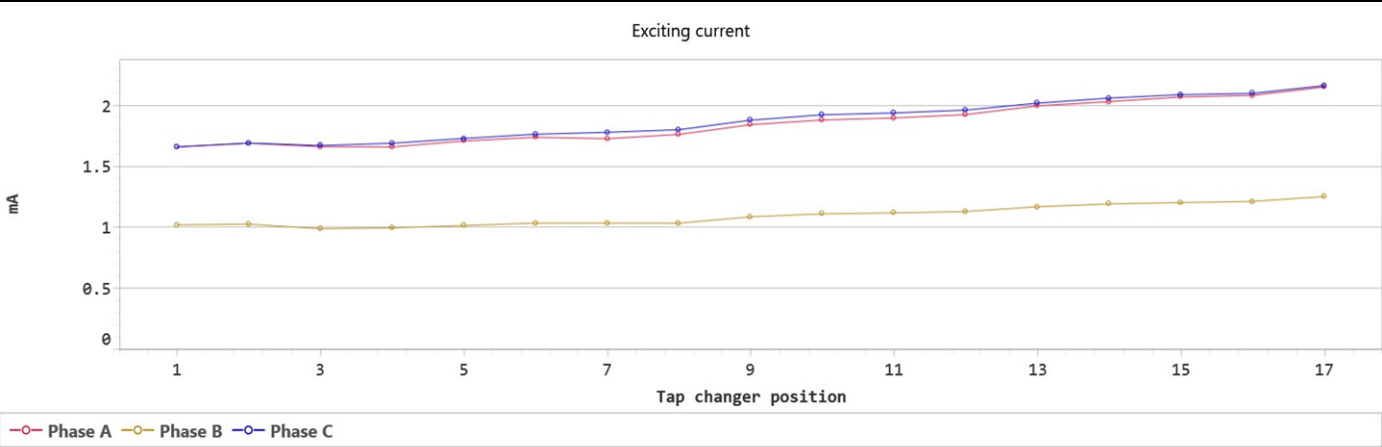
Tap	Phase	Nom. Ratio	$V_{prim} (L-L)$	I_{prim}	$V_{sec} (L-L)$	V_{phase}	TTR*	Ratio dev	Assessment
1	A	5.2924	119.93 V	1.659 mA	13.09 V	30 °	5.2907	-0.03 %	Pass
1	B	5.2924	119.92 V	1.019 mA	13.09 V	30.01 °	5.2901	-0.04 %	Pass
1	C	5.2924	119.92 V	1.660 mA	13.09 V	30.01 °	5.2908	-0.03 %	Pass
2	A	5.2322	119.92 V	1.690 mA	13.23 V	30 °	5.2346	0.04 %	Pass
2	B	5.2322	119.92 V	1.025 mA	13.23 V	30.01 °	5.2340	0.03 %	Pass
2	C	5.2322	119.92 V	1.691 mA	13.23 V	30.01 °	5.2347	0.05 %	Pass
3	A	5.1721	119.92 V	1.662 mA	13.39 V	30 °	5.1705	-0.03 %	Pass
3	B	5.1721	119.92 V	0.988 mA	13.39 V	30.01 °	5.1698	-0.04 %	Pass
3	C	5.1721	119.92 V	1.671 mA	13.39 V	30.01 °	5.1706	-0.03 %	Pass
4	A	5.1120	119.92 V	1.659 mA	13.54 V	30 °	5.1144	0.05 %	Pass
4	B	5.1120	119.92 V	0.994 mA	13.54 V	30.01 °	5.1138	0.04 %	Pass
4	C	5.1120	119.92 V	1.690 mA	13.54 V	30.01 °	5.1145	0.05 %	Pass
5	A	5.0518	119.92 V	1.708 mA	13.71 V	30 °	5.0505	-0.03 %	Pass
5	B	5.0518	119.92 V	1.014 mA	13.71 V	30.01 °	5.0498	-0.04 %	Pass
5	C	5.0518	119.92 V	1.729 mA	13.71 V	30.01 °	5.0505	-0.03 %	Pass
6	A	4.9917	119.92 V	1.738 mA	13.86 V	30 °	4.9944	0.05 %	Pass
6	B	4.9917	119.92 V	1.034 mA	13.86 V	30.01 °	4.9937	0.04 %	Pass
6	C	4.9917	119.92 V	1.764 mA	13.86 V	30.01 °	4.9944	0.05 %	Pass
7	A	4.9315	119.92 V	1.727 mA	14.04 V	30 °	4.9303	-0.02 %	Pass
7	B	4.9315	119.92 V	1.034 mA	14.04 V	30.01 °	4.9296	-0.04 %	Pass
7	C	4.9315	119.92 V	1.779 mA	14.04 V	30.01 °	4.9303	-0.02 %	Pass
8	A	4.8714	119.92 V	1.762 mA	14.20 V	30 °	4.8742	0.06 %	Pass
8	B	4.8714	119.92 V	1.033 mA	14.21 V	30.01 °	4.8735	0.04 %	Pass
8	C	4.8714	119.92 V	1.801 mA	14.20 V	30.01 °	4.8743	0.06 %	Pass
9	A	4.8113	119.92 V	1.842 mA	14.39 V	30 °	4.8103	-0.02 %	Pass
9	B	4.8113	119.92 V	1.086 mA	14.40 V	30.01 °	4.8094	-0.04 %	Pass
9	C	4.8113	119.92 V	1.879 mA	14.39 V	30.01 °	4.8103	-0.02 %	Pass
10	A	4.7511	119.92 V	1.881 mA	14.56 V	30 °	4.7543	0.07 %	Pass
10	B	4.7511	119.92 V	1.110 mA	14.56 V	30.01 °	4.7535	0.05 %	Pass
10	C	4.7511	119.92 V	1.923 mA	14.56 V	30.01 °	4.7543	0.07 %	Pass
11	A	4.6910	119.92 V	1.897 mA	14.76 V	30 °	4.6903	-0.01 %	Pass
11	B	4.6910	119.92 V	1.118 mA	14.76 V	30.01 °	4.6895	-0.03 %	Pass

<i>Tap</i>	<i>Phase</i>	<i>Nom. Ratio</i>	<i>V prim (L-L)</i>	<i>I prim</i>	<i>V sec (L-L)</i>	<i>V phase</i>	<i>TTR*</i>	<i>Ratio dev</i>	<i>Assessment</i>
11	C	4.6910	119.92 V	1.938 mA	14.76 V	30.01 °	4.6903	-0.01 %	Pass
12	A	4.6308	119.92 V	1.925 mA	14.94 V	30 °	4.6342	0.07 %	Pass
12	B	4.6308	119.92 V	1.129 mA	14.94 V	30.01 °	4.6334	0.06 %	Pass
12	C	4.6308	119.92 V	1.962 mA	14.94 V	30.01 °	4.6343	0.07 %	Pass
13	A	4.5707	119.92 V	1.996 mA	15.15 V	30 °	4.5704	-0.01 %	Pass
13	B	4.5707	119.92 V	1.168 mA	15.15 V	30.01 °	4.5694	-0.03 %	Pass
13	C	4.5707	119.92 V	2.019 mA	15.15 V	30.02 °	4.5703	-0.01 %	Pass
14	A	4.5106	119.92 V	2.032 mA	15.34 V	30 °	4.5143	0.08 %	Pass
14	B	4.5106	119.92 V	1.191 mA	15.34 V	30.01 °	4.5134	0.06 %	Pass
14	C	4.5106	119.92 V	2.060 mA	15.34 V	30.02 °	4.5143	0.08 %	Pass
15	A	4.4504	119.92 V	2.069 mA	15.56 V	30 °	4.4502	0.00 %	Pass
15	B	4.4504	119.92 V	1.202 mA	15.56 V	30.01 °	4.4494	-0.02 %	Pass
15	C	4.4504	119.92 V	2.089 mA	15.56 V	30.01 °	4.4502	-0.01 %	Pass
16	A	4.3903	119.92 V	2.083 mA	15.76 V	30 °	4.3942	0.09 %	Pass
16	B	4.3903	119.92 V	1.212 mA	15.76 V	30.01 °	4.3933	0.07 %	Pass
16	C	4.3903	119.92 V	2.100 mA	15.76 V	30.02 °	4.3941	0.09 %	Pass
17	A	4.3301	119.92 V	2.151 mA	15.99 V	30 °	4.3302	0.00 %	Pass
17	B	4.3301	119.92 V	1.252 mA	15.99 V	30.01 °	4.3294	-0.02 %	Pass
17	C	4.3301	119.92 V	2.163 mA	15.99 V	30.02 °	4.3301	0.00 %	Pass

Graphs for standard test



Graph for exciting current



DC Winding Resistance Prim

Comment

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Measurement settings

Test current	33 A
Output mode	33 A @ 56 V
	3 ph.
Measurement	L-N

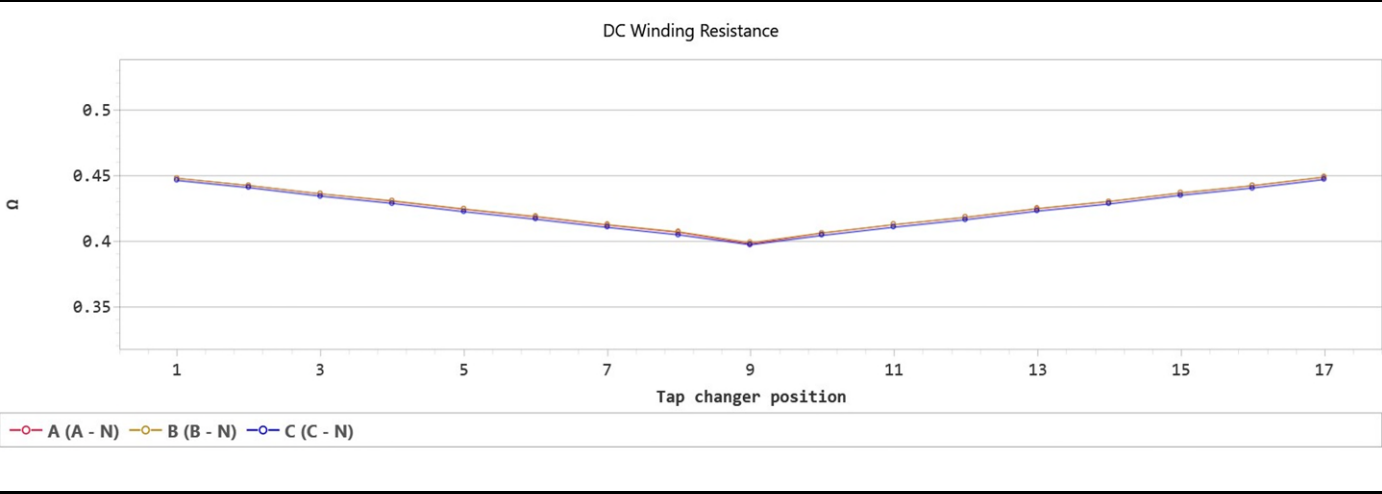
Tap changer under test	OLTC
Automatic tap control	No

Winding temperature	30 °C
Reference temperature	75 °C
Temp. corr. factor (K)	1.17

Automatic result	Yes
Settling time (At)	10 s
Tolerance R dev	0.01 %

	A (A - N)			B (B - N)			C (C - N)		
Tap	R meas	R dev	R corr	R meas	R dev	R corr	R meas	R dev	R corr
1	382.660 mΩ	0.001 %	447.763 mΩ	382.600 mΩ	0.001 %	447.693 mΩ	381.246 mΩ	0.001 %	446.108 mΩ
2	377.934 mΩ	0.001 %	442.233 mΩ	377.998 mΩ	0.000 %	442.307 mΩ	376.561 mΩ	0.000 %	440.626 mΩ
3	372.524 mΩ	0.001 %	435.903 mΩ	372.502 mΩ	0.001 %	435.876 mΩ	371.028 mΩ	0.001 %	434.152 mΩ
4	367.778 mΩ	0.000 %	430.349 mΩ	368.084 mΩ	0.001 %	430.707 mΩ	366.324 mΩ	0.001 %	428.648 mΩ
5	362.444 mΩ	0.001 %	424.107 mΩ	362.717 mΩ	0.002 %	424.427 mΩ	360.893 mΩ	0.001 %	422.293 mΩ
6	357.552 mΩ	0.001 %	418.383 mΩ	357.837 mΩ	0.002 %	418.717 mΩ	355.973 mΩ	0.001 %	416.536 mΩ
7	352.197 mΩ	0.001 %	412.117 mΩ	352.486 mΩ	0.001 %	412.455 mΩ	350.672 mΩ	0.002 %	410.332 mΩ
8	347.400 mΩ	0.001 %	406.504 mΩ	347.792 mΩ	0.002 %	406.963 mΩ	345.792 mΩ	0.001 %	404.623 mΩ
9	339.888 mΩ	0.001 %	397.715 mΩ	340.740 mΩ	0.001 %	398.711 mΩ	339.334 mΩ	0.001 %	397.066 mΩ
10	346.939 mΩ	0.001 %	405.964 mΩ	347.120 mΩ	0.001 %	406.176 mΩ	345.435 mΩ	0.002 %	404.205 mΩ
11	352.423 mΩ	0.001 %	412.381 mΩ	352.538 mΩ	0.002 %	412.516 mΩ	350.781 mΩ	0.001 %	410.460 mΩ
12	357.163 mΩ	0.001 %	417.929 mΩ	357.319 mΩ	0.001 %	418.111 mΩ	355.655 mΩ	0.001 %	416.164 mΩ
13	362.789 mΩ	0.001 %	424.511 mΩ	362.985 mΩ	0.001 %	424.741 mΩ	361.259 mΩ	0.002 %	422.720 mΩ
14	367.498 mΩ	0.002 %	430.022 mΩ	367.642 mΩ	0.001 %	430.189 mΩ	365.924 mΩ	0.001 %	428.180 mΩ
15	373.049 mΩ	0.001 %	436.517 mΩ	373.178 mΩ	0.001 %	436.668 mΩ	371.575 mΩ	0.001 %	434.792 mΩ
16	377.779 mΩ	0.001 %	442.052 mΩ	377.922 mΩ	0.001 %	442.219 mΩ	376.285 mΩ	0.001 %	440.304 mΩ
17	383.336 mΩ	0.002 %	448.553 mΩ	383.492 mΩ	0.001 %	448.736 mΩ	381.870 mΩ	0.001 %	446.838 mΩ

Graphs for standard test



DC Winding Resistance Sec

Comment

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Measurement settings

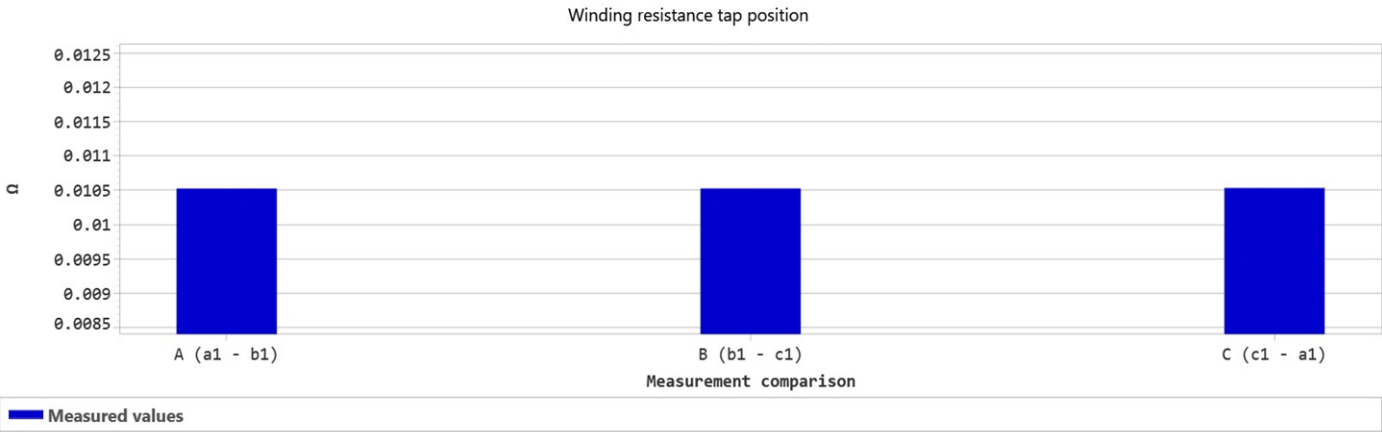
Test current	33 A
Output mode	33 A @ 170 V
	1 ph.

Winding temperature	29 °C
Reference temperature	75 °C
Temp. corr. factor (K)	1.175

Automatic result	Yes
Settling time (At)	10 s
Tolerance R dev	0.01 %

A (a1 - b1)			B (b1 - c1)			C (c1 - a1)		
<i>R meas</i>	<i>R dev</i>	<i>R corr</i>	<i>R meas</i>	<i>R dev</i>	<i>R corr</i>	<i>R meas</i>	<i>R dev</i>	<i>R corr</i>
8.957 mΩ	0.009 %	10.521 mΩ	8.958 mΩ	0.008 %	10.522 mΩ	8.963 mΩ	0.008 %	10.527 mΩ

Graphs for standard test



Demagnetization

Comment

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Test current	5 A
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Saturation level	99 %
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Standard Test

Meas.	Status	I DC	Min. neg. remanence	Max. pos. remanence	Remanence	Initial remanence	Assessment
YNd1	Finished	0.000 A	-477.83 Vs	548.59 Vs	1.03 %	-67.68 %	Man. pass
Pre Excitaion HV Winding	Finished	0.000 A	-454.66 Vs	474.71 Vs	0.46 %	-1.47 %	Man. pass
Final Demag	Finished	0.000 A	-455.39 Vs	488.84 Vs	0.53 %	-65.26 %	Man. pass
Final Demag	Finished	0.000 A	-449.87 Vs	474.58 Vs	0.56 %	0.44 %	Man. pass